

SUPPLEMENT TO: “THE RANDOM PRIORITY MECHANISM IS UNIQUELY SIMPLE, EFFICIENT, AND FAIR”

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This supplement contains proofs of lemmas omitted from the main text.

Proof of Lemma 6

We first show the result for BG lurkers.¹ BG Lemma E.18 shows that the agent who moves at h^N is not a BG lurker at $h^{n'}$ for $n' \leq N$.² Thus, we must show that ℓ does not move at $h^{n'}$ for any $n' = n, n+1, \dots, N-1$. It is sufficient to show the result for the earliest history h^n such that $\ell \in \mathcal{L}^{BG}(h^n)$. Since ℓ first becomes a BG lurker at h^n , we have that ℓ moves at h^{n-1} and: $P_\ell(h^n) \neq G_\ell(h^n)$, $x_\ell \in P_\ell(h^{n-1})$, and $C_\ell^\subseteq(h^{n-1}) = P_\ell(h^{n-1}) \setminus \{x_\ell\}$. Greedy strategies imply that all types $\succ_\ell$ that reach h^n , are such that $Top(\succ_\ell, P_\ell(h^n)) = x_\ell$. Since no agent moves twice in a row, ℓ does not move at h^n . Assume that ℓ moves at $h^{n'}$ for some $n' \in \{n+1, \dots, N-1\}$, and let $h^{n'}$ be the earliest such history where ℓ moves. If $x_\ell \in C_\ell(h^{n'})$, then by greedy strategies, all types of agent ℓ that reach $h^{n'}$ clinch x_ℓ , and thus there is no passing action (it can be pruned) which is a contradiction. If $x_\ell \notin P_\ell(h^{n'})$, we have:³

$$P_\ell(h^{n'}) = P_\ell(h^{n'}) \setminus \{x_\ell\} \subseteq P_\ell(h^{n-1}) \setminus \{x_\ell\} = C_\ell^\subseteq(h^{n-1}) \subseteq C_\ell^{\subseteq \cap} (h^{n'}) \subseteq C_\ell(h^{n'}).$$

In words, since x_ℓ is previously unclinchable and becomes impossible at $h^{n'}$, property (M3) of millipede games implies that the mechanism must offer ℓ at least a clinching action for each object that was previously clinchable, and since ℓ was already a lurker, this includes everything that was previously possible (except for x_ℓ , which is no longer possible). The above implies $P_\ell(h^{n'}) \subseteq C_\ell(h^{n'})$, i.e., everything possible at $h^{n'}$ is also clinchable at $h^{n'}$. By property 4, there is no passing action, again a contradiction. The only remaining possibility is that $x_\ell \in P_\ell(h^{n'}) \setminus C_\ell(h^{n'})$. By the structure of millipede games, x_ℓ can only be possible following the unique passing action at $h^{n'}$. Since $Top(\succ_\ell, P_\ell(h^{n'})) = x_\ell$ for all types of agent ℓ that reach $h^{n'}$, all types of agent ℓ who reach $h^{n'}$ must pass at $h^{n'}$, and since the game is pruned, there are no clinching actions at $h^{n'}$. But then, $h^{n'}$ is a trivial move with $|A(h)| = 1$, which is a contradiction. Thus, for any $\ell \in \mathcal{L}^{BG}(h^n)$, ℓ does not move at $h^{n'}$ for any $n' \in \{n, n+1, \dots, N\}$.

Because $\mathcal{L}(h) \subseteq \mathcal{L}^{BG}(h)$ at any h , that the same result holds for lurkers is immediate. ■

¹Recall that BG refers to [Bade and Gonczarowski \(2017\)](#).

²To be precise, BG Lemma E.18 says if $\mathcal{L}^{BG}(h) \neq \emptyset$, then the agent t who moves at the maximal superhistory of the form $h' = (h, a^*, \dots, a^*)$ is not a BG lurker at h . It is obvious that t is also not a BG lurker at any $\tilde{h}$ such that $\mathcal{L}^{BG}(\tilde{h}) = \emptyset$ (because there are no BG lurkers at $\tilde{h}$).

³The first equality follows because $x_\ell \notin P_\ell(h^{n'})$, the first inclusion because the set of possible objects only shrinks as we move along the game path, the second equality because ℓ is a BG lurker for x_ℓ at h^n , the second inclusion because the set of previously clinchable objects only grows as we move along the game path, and the last inclusion by property (M3) of millipede games.

Proof of Lemma 7

We first show the result for BG lurkers. For $n = 1$, $\mathcal{L}^{BG}(h^{n-1}) = \mathcal{L}^{BG}(h^0) = \emptyset$, and so the statement is obvious. So, assume it holds for all $n' = 1, \dots, n-1$. Take some $\ell \in \mathcal{L}^{BG}(h^{n-1})$ (if there is no such ℓ and so $\mathcal{L}^{BG}(h^{n-1}) = \emptyset$, then the result is obvious). With slight abuse of notation, we interpret ℓ to mean that agent ℓ is the ℓ^{th} oldest BG lurker, and let x_ℓ be the object that ℓ BG lurks. We will show that conditions (i)-(iii) in the definition of a BG-lurker hold at h^n . Notice that for conditions (ii) and (iii), they refer to the most recent move of ℓ strictly prior to h^n . Since $\ell \in \mathcal{L}^{BG}(h^{n-1})$, conditions (ii) and (iii) held at ℓ 's most recent move strictly prior to h^{n-1} , and by Lemma 6, ℓ does not move again at any $h^{n'}$ after becoming a BG-lurker, so conditions (ii) and (iii) continue to hold at h^n . Thus, we must show that condition (i) continues to hold at h^n .

Consider $n < N$. If $P_\ell(h^n) = G_\ell(h^n)$ (i.e., condition (i) fails), then properties 3 and 4 imply there is no passing action at h^n (i.e., h^n is a terminating history), which is a contradiction. Therefore, $P_\ell(h^n) \neq G_\ell(h^n)$, and condition (i) holds.

Now consider $n = N$. Since $\ell \in \mathcal{L}^{BG}(h^{N-1})$, we have $P_\ell(h^{N-1}) \neq G_\ell(h^{N-1})$. We want to show that $P_\ell(h^N) \neq G_\ell(h^N)$. So, assume not, i.e., $P_\ell(h^N) = G_\ell(h^N)$.

CLAIM: *Let $n < N$ and consider a clinching action $a_x \in A(h^n)$ that clinches object x for the agent i that moves at h^n . Then, (i) x is not BG-lurked at h^n and (ii) at (h^n, a_x) , each BG lurker $\ell' \in \mathcal{L}^{BG}(h^n)$ has clinched their lurked object.*

Proof. If $\mathcal{L}^{BG}(h^n) = \emptyset$, then part (i) is obvious and (ii) is vacuous. So, assume $\mathcal{L}^{BG}(h^n) \neq \emptyset$. By Lemma 6, $i \notin \mathcal{L}^{BG}(h^n)$. If $x \in \mathcal{X}^{\mathcal{L}, BG}(h^n)$, by BG Lemma E.14, $P_i(h^n) = G_i(h^n)$, and by properties 3 and 4, there is no passing action at h^n , a contradiction. This establishes part (i). Given part (i), BG Lemma E.17 shows that each BG lurker ℓ' at h^n receives their BG-lurked object at all terminal $\bar{h} \supseteq (h^n, a_x)$, which establishes part (ii). ■

Since $P_\ell(h^{N-1}) \neq G_\ell(h^{N-1})$, it must be that $x_\ell \notin G_\ell(h^{N-1})$.⁴ Since we've assumed $P_\ell(h^N) = G_\ell(h^N)$, this implies that $x_\ell \in P_\ell(h^N)$, and so by obvious dominance, $G_\ell(h^N) = \mathcal{X} \setminus \{x_1, \dots, x_\ell\}$, where $x_1, \dots, x_{\ell-1}$ are the BG-lurked objects older than x_ℓ .⁵

Now, at h^N , the agent who moves, t (the terminator, who is not a BG lurker by BG Lemma E.18) has only clinching actions. Since $G_\ell(h^N) = \mathcal{X} \setminus \{x_1, \dots, x_\ell\}$, none of these objects can be clinchable for t at h^N . In particular, $C_t(h^N) \subseteq \{x_1, \dots, x_\ell\}$. We further claim that $|C_t(h^N)| > 1$. To see this, presume that $|C_t(h^N)| = 1$, and let x' be the unique object that is clinchable at h^N for t . Let history $h^{n'}$ be the most recent history at which t moves strictly prior to h^N . Such a history exists because by BG Lemma E.18, $C_t^\subseteq(h^N) = \mathcal{X}$. Additionally, the fact that no agent moves twice in a row implies that $n' \leq N-2$. Note also that $C_t^\subseteq(h^{n'}) \neq \mathcal{X}$ (as otherwise,

⁴This follows because if x_ℓ is guaranteeable at h^N , then ℓ has effectively already clinched it prior to h^N , which means she was not a BG lurker. More formally, assume $x_\ell \in G_\ell(h^N)$, and let h^* be the earliest history at which ℓ is a BG-lurker for x_ℓ ; since $\ell \in \mathcal{L}^{BG}(h^{N-1})$, we have $h^* \subsetneq h^N$. By definition of a BG lurker and greedy strategies, $Top(>_\ell, P_\ell(h^*)) = x_\ell$ for any type that reaches h^* . By the claim just proven, ℓ has clinched x_ℓ following any clinching action at any (h^n, a_x) , where $h^* \subseteq h^n \subseteq h^{N-1}$. This means that if $x_\ell \in G_\ell(h^N)$, then x_ℓ is already guaranteeable at h^* : $x_\ell \in G_\ell(h^*)$. Further, no type $>_\ell$ that reaches h^* will follow any strategy for which anything but x_ℓ is a possible outcome, and so by pruning, $P_\ell(h^*) = G_\ell(h^*)$. But this contradicts that ℓ is a lurker at h^* .

⁵BG Lemma E.16 implies that $P_\ell(h^{N-1}) = \mathcal{X} \setminus \{x_1, \dots, x_{\ell-1}\}$ (note that $x_\ell \in P_\ell(h^{N-1})$), and also $C_\ell^\subseteq(h^{N-1}) = P_\ell(h^{N-1}) \setminus \{x_\ell\}$. If there is some $x \in \mathcal{X} \setminus \{x_1, \dots, x_\ell\}$ such that $x \notin G_\ell(h^N)$, then the type that ranks $>_\ell: x_\ell, x, \dots$ has no obviously dominant strategy: because x_ℓ is possible at h^{N-1} , but has never been clinchable, any obviously dominant strategy requires ℓ to pass at all $h' \subseteq h^{N-1}$ where she is called to move. Since $x_\ell \notin P_\ell(h^N)$, and $x \notin G_\ell(h^N)$, any such strategy has a worst-case outcome that is strictly worse than x . As x was clinchable for ℓ at some $\tilde{h} \subsetneq h^{N-1}$, no such strategy can be obviously dominant.

by greedy strategies, all types of agent t will have clinched an object by $h^{n'}$, and thus by pruning, the passing path will not continue to h^N , which is a contradiction). Therefore, it must be $C_t^{\subseteq}(h^{n'}) = \mathcal{X} \setminus \{x'\}$. Greedy strategies imply that all types of agent t that reach $h^{n'+1}$ rank object x' first (among all objects). Consider the histories $h^{n'+1}, \dots, h^{N-1}$, and notice that, by definition of $h^{n'}$, agent t does not move at any of these histories.

There are two cases: (i) x' is not clinchable at any $\tilde{h} \in \{h^{n'+1}, \dots, h^{N-1}\}$ or (ii) x' is clinchable at at least one such $\tilde{h} \in \{h^{n'+1}, \dots, h^{N-1}\}$. In case (i), for all $\tilde{h} \in \{h^{n'+1}, \dots, h^{N-1}\}$, following any clinching action $(\tilde{h}, a_z)$, there is some $z \neq x'$ that is clinched. Because the game is pruned and $C_t^{\subseteq}(h^{n'}) = \mathcal{X} \setminus \{x'\}$, obvious dominance requires that agent t receive x' at any terminal history $\bar{h} \supseteq (\tilde{h}, a_z)$. Further, t receives x' at any terminal $\bar{h} \supseteq h^N$ as well, which implies that t receives x' at all terminal $\bar{h} \supseteq h^{n'+1} = (h^{n'}, a^*)$. This means that t has clinched x' at $h^{n'+1}$ already, which contradicts that there is a passing action at $h^{n'}$. In case (ii), since x' is clinchable at $\tilde{h}$ for an agent other than t , and all other objects $\mathcal{X} \setminus \{x'\}$ were previously clinchable for t , obvious dominance requires that following this clinching action, agent t must receive $Top(>, \mathcal{X} \setminus \{x'\})$; in particular, every object in $\mathcal{X} \setminus \{x'\}$ must remain possible at $h^{n'+1}$ for t . Object x' itself is also still possible, so we have $P_t(h^{n'+1}) = \mathcal{X}$.

This implies that conditions (ii) and (iii) in the definition of a BG lurker for x' hold at $h^{n'+1}$.⁶ Furthermore, if $P_t(h^{n'+1}) = G_t(h^{n'+1})$, then by properties 3 and 4 of millipede games, there is no passing action at $h^{n'+1}$, and as $n' \leq N-2$, this a contradiction. Thus, $P_t(h^{n'+1}) \neq G_t(h^{n'+1})$, and condition (i) of a BG lurker holds as well. Thus, t is a BG lurker at $h^{n'+1}$. However, this is a contradiction to BG Lemma E.18 (cf. footnote 2).

Thus, we may assume that $|C_t(h^N)| > 1$. Since we have also shown that $C_t(h^N) \subseteq \{x_1, \dots, x_\ell\}$, at least two objects in $\{x_1, \dots, x_\ell\}$ must be clinchable at h^N (this also implies that $\ell \geq 2$ in the present context). Choose any two of these objects, and denote them $x_{\ell'}$ and $x_{\ell''}$, where without loss of generality, $1 \leq \ell' < \ell'' \leq \ell$. These are both BG-lurked objects, and their BG lurkers are agents ℓ' , ℓ'' . If t clinches $x_{\ell'}$, at $(h^N, a_{x_{\ell'}})$, agent ℓ' must be able to guarantee all objects in $\mathcal{X} \setminus \{x_1, \dots, x_{\ell'}\}$, and agent ℓ must be able to guarantee all objects in $\mathcal{X} \setminus \{x_1, \dots, x_\ell\}$, which is a contradiction.⁷

Therefore, $P_\ell(h^N) \neq G_\ell(h^N)$, condition (i) holds, and $\ell \in \mathcal{L}^{BG}(h^N)$.

Thus, we have shown that $\mathcal{L}^{BG}(h^{n-1}) \subseteq \mathcal{L}^{BG}(h^n)$ for all $n = 1, \dots, N$. To see that the same is true for lurkers, take some $\ell \in \mathcal{L}(h^{n-1})$. Notice that $\mathcal{L}(h^{n-1}) \subseteq \mathcal{L}^{BG}(h^{n-1})$, and so $\ell \in \mathcal{L}^{BG}(h^{n-1})$, as we just showed above. That is, ℓ satisfies conditions (i)-(iii) in the definition. So all that is left to show is that condition (iv) holds at h^n . But, notice that similar to conditions (ii) and (iii), condition (iv) refers only to ℓ 's most recent move strictly prior to h^n . As $\ell \in \mathcal{L}(h^{n-1})$, condition (iv) holds at h^{n-1} , and by Lemma 6, ℓ does not move at any $h^{n'}$ after becoming a lurker for the first time. Thus, condition (iv) continues to hold at h^n , and $\ell \in \mathcal{L}(h^n)$. ■

⁶Notice that the conditions refer to t 's most recent move strictly prior to $h^{n'+1}$, which, by definition is $h^{n'}$. Since $P_t(h^{n'+1}) = \mathcal{X}$, we have $P_t(h^{n'}) = \mathcal{X}$ as well, and it is obvious that $x' \in P_t(h^{n'})$, which is condition (ii). Condition (iii) says $C_t^{\subseteq}(h^{n'}) = P_t(h^{n'}) \setminus \{x'\}$, which we showed two paragraphs above.

⁷This follows an argument similar to footnote 5: when $x_{\ell'}$ becomes impossible for ℓ' at $(h^N, a_{x_{\ell'}})$, obvious dominance requires that ℓ' be able to guarantee everything that was previously clinchable. By BG Lemma E.16 and the fact that ℓ' is a BG lurker for $x_{\ell'}$, this is all objects in $\mathcal{X} \setminus \{x_1, \dots, x_{\ell'}\}$. (Note that it is impossible for all objects to be BG-lurked, because as BG Lemma E.18 shows, the terminator is not a BG lurker. Therefore, $\{x_1, \dots, x_{\ell'}\} \not\subseteq \mathcal{X}$, and so $(\mathcal{X} \setminus \{x_1, \dots, x_{\ell'}\}) \cap (\mathcal{X} \setminus \{x_1, \dots, x_\ell\}) \neq \emptyset$.)

Proof of Lemma 8

By BG Lemma E.14, we have $P_j(h) = G_j(h)$. By properties 3 and 4, there is no passing action at h , and thus, h is a terminating history, which establishes the first statement.

We must also show that j moves at h . Assume some $k \neq j$ moved at h , and thus k is the terminator. Let $\tilde{h} \subseteq h$ be the smallest subhistory of h such that x is BG-lurked at $\tilde{h}$ (i.e., it is not BG-lurked at any predecessor of $\tilde{h}$). We claim that $x \in P_j(\tilde{h})$. To see this, assume $x \notin P_j(\tilde{h})$. Since $x \in C_j^\subseteq(h)$, we must have $x \in C_j(\hat{h})$ for some $\hat{h} \not\subseteq \tilde{h} \subseteq h$. This implies that both j was active strictly prior to $\tilde{h}$, and also $\tilde{h} = h$.⁸ This further implies that k is not active strictly before h .⁹ By BG Lemma E.18, $C_k^\subseteq(h) = \mathcal{X}$, and since h is the first time k is active, $C_k(h) = \mathcal{X}$, i.e., k can clinch every object at h . But, this means that at h , no object can be guaranteeable for j , which contradicts $P_j(h) = G_j(h)$. Therefore, $x \in P_j(\tilde{h})$. BG Lemma E.18 then implies that j is the terminator, and thus j moves at h . ■

Proof of Lemma 9

The statement is obvious for $n = 0$. So, presume it holds for $1, \dots, n-1$, where $n < N$. We show that $\mathcal{L}(h^n) = \mathcal{L}^{BG}(h^n)$.

Notice that we can write $\mathcal{L}^{BG}(h^{n-1}) = \mathcal{L}(h^{n-1}) \subseteq \mathcal{L}(h^n) \subseteq \mathcal{L}^{BG}(h^n)$, where the first equality is the inductive hypothesis, the first inclusion is Lemma 7, and the second inclusion is because at any history h , any lurker is also a BG lurker. Thus, if $\mathcal{L}^{BG}(h^{n-1}) = \mathcal{L}^{BG}(h^n)$, then the inclusions must be equalities, and we are done.

So, consider the case that $\mathcal{L}^{BG}(h^{n-1}) \neq \mathcal{L}^{BG}(h^n)$, and assume that $\mathcal{L}(h^n) \neq \mathcal{L}^{BG}(h^n)$. As $\mathcal{L}^{BG}(h^{n-1}) \subseteq \mathcal{L}^{BG}(h^n)$ and $h^n = (h^{n-1}, a^*)$, it must be that there is some agent i who first becomes a BG lurker at h^n , and i must move at h^{n-1} .¹⁰ In particular, $\mathcal{L}^{BG}(h^n) = \mathcal{L}^{BG}(h^{n-1}) \cup \{i\}$. As $\mathcal{L}^{BG}(h^{n-1}) = \mathcal{L}(h^{n-1}) \subseteq \mathcal{L}(h^n) \subseteq \mathcal{L}^{BG}(h^n) = \mathcal{L}^{BG}(h^{n-1}) \cup \{i\}$, if $\mathcal{L}(h^n) \neq \mathcal{L}^{BG}(h^n)$, then it must be that $i \notin \mathcal{L}(h^n)$. Since $i \in \mathcal{L}^{BG}(h^n)$, conditions (i)-(iii) hold at h^n for i , and thus, it must be condition (iv) that is violated, i.e., at h^{n-1} , there is some active non-lurker j at h^{n-1} such that $x \in C_j^\subseteq(h^{n-1})$ (where x is the object that i BG-lurks). Since $\mathcal{L}^{BG}(h^{n-1}) = \mathcal{L}(h^{n-1})$, agent j is also a BG nonlurker at h^{n-1} , and thus, by Lemma 8, there is no passing action at h^n , a contradiction. ■

Proof of Lemma 10

As above, let $h^n = (h_\emptyset, a^*, \dots, a^*)$, where h^N is the terminating history. Lemma 9 shows that at h^n for $n < N$, we have $\mathcal{L}^{BG}(h^n) = \mathcal{L}(h^n)$. The only remaining history to consider is the terminating history h^N . Note that $\mathcal{L}^{BG}(h^{N-1}) = \mathcal{L}(h^{N-1}) \subseteq \mathcal{L}(h^N) \subseteq \mathcal{L}^{BG}(h^N)$, where the first equality follows from Lemma 9, the first inclusion by Lemma 7, and the second

⁸As j is a BG non-lurker at h , she is also a BG non-lurker at $\hat{h}$ and $\tilde{h}$, by Lemma 7. Similar to the paragraph above, by BG Lemma E.14, $P_j(\tilde{h}) = G_j(\tilde{h})$, and so $\tilde{h}$ is the terminating history, i.e., $\tilde{h} = h$.

⁹Let h' be the immediate predecessor of h , so $h = (h', a^*)$, and let i be the agent who BG-lurks x . Since x became BG-lurked only at the terminating history h , agents i and j are active BG non-lurkers at h' , and by BG Lemma E.19, there can be at most two active BG non-lurkers at any history. Therefore, at h' , k is either inactive, or a BG lurker. BG Lemma E.18 shows that the terminator is not a BG lurker at any $\tilde{h} \subseteq h$ (see footnote 2), and thus k must become active only at h .

¹⁰For any j that does not move at h^{n-1} , then her most recent move was at some strict subhistory $h' \subsetneq h^{n-1}$. Conditions (ii)-(iii) are thus the same when checking whether j is a BG lurker at h^{n-1} and h^n , so if j is not a BG lurker at h^{n-1} , it must be condition (i) that is violated, i.e., $P_j(h^{n-1}) = G_j(h^{n-1})$. But, by properties 3 and 4, this means there is no passing action at h^{n-1} , a contradiction.

inclusion by the fact that any lurker is a BG lurker. If $\mathcal{L}^{BG}(h^{N-1}) = \mathcal{L}^{BG}(h^N)$, then the inclusions must be equalities, and so all BG lurkers are lurkers. If $\mathcal{L}^{BG}(h^{N-1}) \neq \mathcal{L}^{BG}(h^N)$, it must be that $\mathcal{L}^{BG}(h^N) = \mathcal{L}^{BG}(h^{N-1}) \cup \{i\}$ for some i who first becomes a BG lurker at h^N and also i moves at h^{N-1} (see footnote 10). So, we have $\mathcal{L}^{BG}(h^{N-1}) = \mathcal{L}(h^{N-1}) \subseteq \mathcal{L}(h^N) \subseteq \mathcal{L}^{BG}(h^N) = \mathcal{L}^{BG}(h^{N-1}) \cup \{i\}$. If $i \in \mathcal{L}(h^N)$, then $\mathcal{L}(h^N) = \mathcal{L}^{BG}(h^N)$. If $i \notin \mathcal{L}(h^N)$, then $\mathcal{L}(h^N) = \mathcal{L}^{BG}(h^N) \setminus \{i\}$. In either case, there is at most one BG lurker that is not a lurker. Notice also that we have shown that $\mathcal{L}(h^n) \neq \mathcal{L}^{BG}(h^n)$ only if $n = N$, i.e., this occurs only at the terminating history. Further, the above argument establishes that when this occurs, it is the youngest BG lurker (agent i) who is not a lurker, and i moves at h^{N-1} .

For statement (iv) and (v), since h is the terminating history, the agent who moves at h is the terminator t , who is not a BG lurker (BG Lemma E.18), and thus is also not a lurker. To see that the active agents are $\mathcal{L}^{BG}(h) \cup \{t\}$, note that when i became a BG lurker but not a lurker at $h^N (= h)$, it must be because there was some other j who was an active non-lurker at h^{N-1} and $x \in C_j^\subseteq(h^{N-1})$, where x is the object that i BG-lurks. At h^{N-1} , $\mathcal{L}^{BG}(h^{N-1}) = \mathcal{L}(h^{N-1})$, which also means that all non-lurkers are also BG non-lurkers at h^{N-1} , and so j is a BG non-lurker at h^{N-1} . Since there can be at most two active BG non-lurkers at any history (BG Lemma E.19), at h^{N-1} , the active agents are $\mathcal{L}^{BG}(h^{N-1}) \cup \{i, j\}$. Agent i moves at h^{N-1} , and by Lemma 8, j is the terminator, i.e., $j = t$ and so j moves at h^N . Therefore, at h^N , i becomes a BG lurker and no new agent enters the game, so the active agents at h^N are $\mathcal{L}^{BG}(h^N) \cup \{t\} = \mathcal{L}(h^N) \cup \{i, t\}$. ■

Proof of Lemma 13

For the first part, for any $x \in \bar{\mathcal{X}}^\mathcal{L}(h)$ that is also BG-unlurked, the statement follows from BG Lemma E.11. So, consider some $x \in \bar{\mathcal{X}}^\mathcal{L}(h)$ but $x \in \mathcal{X}^{\mathcal{L}, BG}(h)$. By Lemma 10, there is only one such object, and it is $x = x_{\lambda_{BG}(h)}$, the youngest BG-lurked object at h . Further, this condition only obtains when h is a terminating history, and the active agents at h are $\ell_1, \dots, \ell_{\lambda_{BG}(h)}, t$ where: $\ell_1, \dots, \ell_{\lambda_{BG}(h)-1}$ are both lurkers and BG lurkers, $\ell_{\lambda_{BG}(h)}$ is a BG lurker but not a lurker, and t is the terminator (and neither a lurker nor a BG lurker). By BG Lemma E.16, $x_{\lambda_{BG}(h)} \in P_{\ell'}(h)$ for all $\ell' \in \{\ell_1, \dots, \ell_{\lambda_{BG}(h)}\}$, while by BG Lemma E.18, $x_{\lambda_{BG}(h)} \in C_t^\subseteq(h)$.

The second part follows from the first part and the definition of a lurker. ■

Proof of Lemma 14

If x_ℓ is lurked at h then x_ℓ is BG-lurked at h ; thus if i is a BG non-lurker at h , then the result follows from Lemma 8. So, assume that i is a non-lurker that is a BG lurker at h . We claim that for any lurked object $x_\ell \in \bar{\mathcal{X}}^\mathcal{L}(h)$, we have $x_\ell \notin C_i^\subseteq(h)$, and so the result holds vacuously. To show it, let h' be such that $h = (h', a^*)$, i.e., h' is the immediate predecessor of h . By Lemma 10, h must be a terminating history, agent i moves at h' and passes, and becomes a BG lurker at h . Note that since x_ℓ is lurked at h , it is also BG-lurked at h . If $x_\ell \in C_i^\subseteq(h)$, then, since i does not move at h , we have $x_\ell \in C_i^\subseteq(h')$ as well. Further, x_ℓ cannot be the object i BG lurks at h (because it has been previously clinchable for her), and so x_ℓ must be BG-lurked at h' by some other agent. But then, at h' , i is not a BG lurker, and $x_\ell \in C_i^\subseteq(h')$ for some BG-lurked object x_ℓ . Thus, by Lemma 8, h' is a terminating history, which is a contradiction. ■

Proof of Lemma 15

Since there is a lurked object at h , there is also a BG-lurked object at h . Parts (ii) and (iii) then follow immediately from the corresponding statements in BG Lemmas E.18 and E.20.

BG Lemma E.18 part (i) says that agent t is not a BG lurker at h , and thus, agent t is not a lurker either, which shows part (i). What remains is to show part (iv). For all $h \not\subseteq h'$, any non-lurker is also a BG non-lurker by Lemma 10, and any lurked object is also a BG lurked object, and so the result follows from the corresponding lemma of BG. Thus, consider h' . By Lemma 10, at h' , either $\mathcal{L}^{BG}(h') = \mathcal{L}(h')$ or $\mathcal{L}(h') = \mathcal{L}^{BG}(h') \setminus \{\ell_{\lambda^{BG}(h')}\}$. Similarly, either $\mathcal{X}^{\mathcal{L},BG}(h') = \mathcal{X}^{\mathcal{L}}(h')$ or $\mathcal{X}^{\mathcal{L}}(h') = \mathcal{X}^{\mathcal{L},BG}(h') \setminus \{x_{\lambda^{BG}(h')}\}$. If j is a BG non-lurker, then the result is immediate from the corresponding lemma of BG. It remains to consider j who is a non-lurker but a BG lurker. By Lemma 10, j is a BG lurker for $x_{\lambda^{BG}(h')}$. Notice that $x_{\lambda^{BG}(h')}$ is not lurked at h' (though it is BG-lurked). Thus, the lurked objects at h' are $\mathcal{X}^{\mathcal{L}}(h') = \{x_1, \dots, x_{\lambda^{BG}(h')-1}\}$. By Lemma E.16 from BG, $P_j(h') = \mathcal{X}(h') \setminus \{x_1, \dots, x_{\lambda^{BG}(h')-1}\}$; in other words, for any $x \in \mathcal{X}^{\mathcal{L}}(h')$, we have $x \notin P_j(h')$, and so the statement holds vacuously. ■

Proof of Lemma 17

At h , j is a non-lurker, but may or may not be a BG lurker. First consider the case that j is a BG lurker at h , and let y be the object that j BG-lurks. By Lemma 10, j is the youngest BG-lurker, and the set of BG-lurked objects is $\mathcal{X}^{\mathcal{L},BG}(h) = \mathcal{X}^{\mathcal{L}}(h) \cup \{y\}$, where $y \notin \mathcal{X}^{\mathcal{L}}(h)$ and y is the youngest BG-lurked object at h . In addition, the set of unlurked objects is $\bar{\mathcal{X}}^{\mathcal{L},BG}(h) = \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{y\}$. Because j BG-lurks y , we have $y = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h))$. If $y = \hat{x}$, then BG Lemma E.17 implies that j receives $\text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L},BG}(h))$, and further, $y = \hat{x}$ implies $\bar{\mathcal{X}}^{\mathcal{L},BG}(h) = \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\}$, and so the statement holds. If $y \neq \hat{x}$, then BG Lemma E.17 implies that j receives y . Since $y \neq \hat{x}$ and $\hat{x} \in \bar{\mathcal{X}}^{\mathcal{L}}(h)$, we have $y = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h)) = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\})$, and again, the statement holds.

Second, consider the case that j is not a BG lurker at h . In this case, we claim that $\mathcal{L}(h) = \mathcal{L}^{BG}(h)$. Indeed, the agent who moves at h , agent i , is not a lurker nor a BG lurker (Lemma 6), and so both i and j are both non-lurkers and BG non-lurkers at h . As there can be at most 2 non-lurkers at any h (Lemma 12) and the same is true for BG non-lurkers (BG Lemma E.19), all other active agents at h are both lurkers and BG lurkers. Similar statements apply to non-lurkers ($\bar{\mathcal{L}}(h) = \bar{\mathcal{L}}^{BG}(h)$), lurked objects ($\mathcal{X}^{\mathcal{L}}(h) = \mathcal{X}^{\mathcal{L},BG}(h)$), and unlurked objects ($\bar{\mathcal{X}}^{\mathcal{L}}(h) = \bar{\mathcal{X}}^{\mathcal{L},BG}(h)$).

We next claim that $P_j(h) \subseteq \bar{\mathcal{X}}^{\mathcal{L}}(h)$. Indeed, if there was some lurked object $x_\ell \in \mathcal{X}^{\mathcal{L}}(h)$ such that $x_\ell \in P_j(h)$, then by Lemma 15, part (iv), j is the terminator. This further implies that the agent i who moves at h is not the terminator, and again by Lemma 15, part (iv), object x (the object i clinches at h) is unlurked at h . This means that $\hat{x} = x$, by definition of $\hat{x}$. However, because j is the terminator, Lemma 15, part (iii) implies $C_i(h) \cap C_j^c(h) = \emptyset$, and in particular, $\hat{x} \notin C_i(h)$, which is a contradiction. Therefore, $P_j(h) \subseteq \bar{\mathcal{X}}^{\mathcal{L}}(h)$.

Let $y = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h))$. Consider first the case $y \notin C_j^c(h)$, and note that this implies that $y \neq \hat{x}$. Because j does not move at h , $C_j^c(h) = C_j^c(h)$ and hence $y \notin C_j^c(h)$. As $\bar{\mathcal{X}}^{\mathcal{L}}(h) = \bar{\mathcal{X}}^{\mathcal{L},BG}(h)$ and BG Lemma E.11 tells us that $\bar{\mathcal{X}}^{\mathcal{L},BG}(h) \subseteq P_j(h) \cup C_j^c(h)$, we conclude that $y \in P_j(h)$. Property (M3) of millipede games then implies that j receives y .¹¹ Since $y \neq \hat{x}$ and

¹¹If j does not receive y , then at some point on the path from h to $\bar{h}$, y became impossible for j before being clinchable (or else by greedy strategies, j would have clinched y , which follows from the definition of y as $\text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h))$ and the fact that $y \in P_j(h) \subseteq \bar{\mathcal{X}}^{\mathcal{L}}(h)$). Property (M3) requires that agent j then either receives or is reoffered an opportunity to clinch each object that was previously clinchable, which includes $\hat{x}$. However, $\hat{x}$ is clinched by another agent before j might be called to move following h (by Lemma 16), and so this is a contradiction. Therefore, y cannot become impossible for j before being clinchable, and as it is her top still-possible choice, when it is offered to her, she will clinch it.

$\hat{x} \in \bar{\mathcal{X}}^{\mathcal{L}}(h)$, we have $y = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h)) = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\})$, and the claim of the lemma obtains.

It remains to consider the case $y \in C_j^{\subseteq}(h)$. Let z be the object j receives at $\bar{h}$, and note that $z \in \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\}$ ($z \in \bar{\mathcal{X}}^{\mathcal{L}}(h)$ follows from $P_j(h) \subseteq \bar{\mathcal{X}}^{\mathcal{L}}(h)$, and $z \neq \hat{x}$ follows because $\hat{x}$ is assigned to another agent at $\bar{h}$, by Lemma 16). Obvious dominance implies that $z \succeq_j y$; otherwise, type $>_j$'s strategy would not obviously dominate the alternative strategy that clinches y at the $h' \not\subseteq h$ where it was offered to her for clinching. Since $z \in \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\}$ and $z \succeq_j y = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h))$, we must have $z = y$ and $y = \text{Top}(>_j, \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\})$, which completes the proof.¹² ■

Proof of Lemma 18

We first claim that there exists some $y \in C_j^{\subseteq}(h)$ such that $y \in \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\}$. This follows because by supposition j moves prior to h , and so must be offered some object y for clinching when she moved. Lemma 14 implies $y \notin \mathcal{X}^{\mathcal{L}}(h)$, and by supposition of the current lemma, $\hat{x} \notin C_j^{\subseteq}(h)$. Therefore, we must have $y \in \bar{\mathcal{X}}^{\mathcal{L}}(h) \setminus \{\hat{x}\}$.

Next, we claim that $C_j^{\subseteq}(h) \subseteq G_j(\hat{h})$.¹³ To see this, take an arbitrary $y \in C_j^{\subseteq}(h)$; as we just showed, such y exists. As $\hat{x} \notin C_j^{\subseteq}(h)$, we know that $y \neq \hat{x}$ and hence a type $>_j: \hat{x}, y, \dots$ is well defined. As $\hat{x} \notin C_j^{\subseteq}(h)$, Lemma 13 implies that $\hat{x} \in P_j(h)$. Because $\hat{x} \in P_j(h) \setminus C_j^{\subseteq}(h)$, greedy strategies imply that type $>_j$ must pass at all $h' \not\subseteq h$ where j moves, and as j does not move between h and $\hat{h}$, this type reaches $\hat{h}$. If $y \notin G_j(\hat{h})$, then, for any strategy type $>_j$ follows starting from $\hat{h}$, type $>_j$ might end up with an object strictly worse than y (because $\hat{x}$ has been clinched by another agent, and y is not guaranteeable at $\hat{h}$). However, j could have clinched y at some history $h' \not\subseteq h$, which contradicts that her strategy is obviously dominant. Thus, $y \in G_j(\hat{h})$ and, as y was an arbitrary element of $C_j^{\subseteq}(h)$, we conclude that $G_j(\hat{h}) \supseteq C_j^{\subseteq}(h)$. The procedure from the proof of the millipede theorem of [Pycia and Troyan \(2023\)](#) allows us to transform the millipede, while maintaining properties 1-5, so that j moves at $\hat{h}$ and any object that is guaranteeable for agent j at history $\hat{h}$ is clinchable at $\hat{h}$, so that $G_j(\hat{h}) = C_j(\hat{h})$. As we just showed that $G_j(\hat{h}) \supseteq C_j^{\subseteq}(h)$, we conclude that $C_j^{\subseteq}(h) \subseteq C_j(\hat{h})$. At $\hat{h}$, agent j may also be offered to clinch additional objects, and may or may not have a passing action. ■

Proof of Lemma 19

First, consider the case that there are no lurked objects at h , and so there are also no lurked objects at any h' such that $h_{\emptyset}^k \subseteq h' \subseteq h$, by Lemma 7. There is then a history h' such that $h_{\emptyset}^k \subseteq h' \not\subseteq h$ and $y \in C_j(h')$. Indeed, if j was not active until the current round, then the existence of h' follows from the assumption that $y \in C_j^{\subseteq}(h)$. If j was active in a previous round then there are two subcases: if j was offered y in a previous round, then j moves at $h_{\emptyset}^k$ and $y \in C_j(h_{\emptyset}^k)$, and we can set $h' = h_{\emptyset}^k$; if j wasn't offered y in a previous round, then there must be some h' in the current round where she was offered it. Hence, history h' is well defined. Since type $>_j$ reaches h , this type must pass at h' . By greedy strategies, there must be some $x \in P_j(h')$ such that $x >_j y$. As there are no lurkers (or lurked objects) at h' or h , $x \in \bar{\mathcal{X}}^{\mathcal{L}}(h') = \bar{\mathcal{X}}^{\mathcal{L}}(h)$. Thus, $\bar{x} >_j y$.

¹²Notice that the above proof continues to apply when there are no lurked objects at h . In this case, the only active agents at h are i and j , and $\hat{x} = x$. When i clinches $\hat{x}$ such that $\hat{x} \in C_j^{\subseteq}(h)$, then j still receives her top choice among all remaining unlurked objects, which in this case is all objects except for $\hat{x}$.

¹³Note that $C_j^{\subseteq}(\hat{h}) = C_j^{\subseteq}(h)$ because j does not move at any h' such that $h \subseteq h' \not\subseteq \hat{h}$.

Now, consider the case where there are lurked objects (and lurkers) at h . By Lemma 15, part (iii), agent j cannot be the terminator. Since j is not the terminator, Lemma 15, part (iv) implies that $P_j(h) \subseteq \bar{\mathcal{X}}^{\mathcal{L}}(h)$. Since i can clinch y at h , there must be some $x \in P_j(h)$ such that $x \succ_j y$, by obvious strategy-proofness; indeed, otherwise, when i clinches y , type $\succ_j$ will end up with an object strictly worse than y , and so the worst-case outcome is strictly worse than the outcome of the alternative strategy of j clinching y at the $h' \not\subseteq h$ where it was clinchable for j . Since $P_j(h) \subseteq \bar{\mathcal{X}}^{\mathcal{L}}(h)$, we have $x \in \bar{\mathcal{X}}^{\mathcal{L}}(h)$, i.e., $\bar{x} \succ_j y$. ■

Proof of Lemma 20

We start with the following lemma.

LEMMA 25: *Assume that there exists an arm α as defined in the statement of Lemma 5, and such that $n \geq 2$. Further, assume that along the initial passing path of the game form, the first lurked objects are (in order) $x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n-1}}$. Then:*

- (a) *For all $n' = 1, \dots, n-1$, the agent that moves at $h_{n'}^*$ in $\Gamma_{n'}$ is agent i , and at $h_{n'}^*$, the number of lurked objects is $P + n' - 1$.*
- (b) *$h_1^* \not\subseteq h_2^* \not\subseteq \dots \not\subseteq h_{n-1}^* \not\subseteq h_n^*$.*
- (c) *For all $n' = 1, \dots, n$, the number of lurked objects at $h_{n'}^*$ is $P + n' - 1$.*
- (d) *For all $n' = 1, \dots, n-1$, $p = 1, \dots, P$, and $n'' = 1, \dots, n'$, in $\Gamma_{n'}$, agent j_p is in the role that lurks x_{j_p} and agent $k_{n''}$ is in the role that lurks $x_{k_{n''}}$.*
- (e) *$h_{n-1}^* \not\subseteq h_{n+1}^*$ and the number of lurked objects at h_{n+1}^* is at least $P + n - 1$.*

Proof of Lemma 25. Part (a). Let $\lambda_{n'}^*$ be the number of lurked objects at history $h_{n'}^*$. Notice that since $\succ_{n'}$ has a tie in the $(P + n')^{th}$ place, we have $\lambda_{n'}^* \leq P + n' - 1$ for all $n' = 1, \dots, n$. Towards a contradiction, assume there was a game $\Gamma_{n'}$ for which i does not move at $h_{n'}^*$. Since $\lambda_{n'}^* \leq P + n' - 1$, the structure of $\succ_{n'}$ implies that the lurked objects are $\{x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{\lambda_{n'}^* - P}}\}$,¹⁴ and the agents coded in step 1 of $\Gamma_{n'}$ are $\{j_1, \dots, j_P, k_1, \dots, k_{\lambda_{n'}^* - P + 1}\}$ (if $\lambda_{n'}^* < P + n' - 1$) or $\{j_1, \dots, j_P, k_1, \dots, k_{\lambda_{n'}^* - P + 1}, i\}$ (if $\lambda_{n'}^* = P + n' - 1$) and the set of lurked objects is $\{x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{\lambda_{n'}^* - P}}\}$. Now, notice that it cannot be a lurked object that is clinched at $h_{n'}^*$. Indeed, if this were true, then $h_{n'}^*$ is the terminating history, which implies that $x_{k_{\lambda_{n'}^* - P}}$ is the last lurked object on the initial passing path of the game (Lemma 14). But, this contradicts the assumption that $x_{k_{\lambda_{n'}^* - P + 1}}$ is the next lurked object on the initial passing path, where notice that such an object exists because $\lambda_{n'}^* - P + 1 \leq n' \leq n - 1$. Thus, it must be an unlurked object that is clinched at $h_{n'}^*$. In particular, by the structure of $\succ_{n'}$, the only possibilities are that agent $k_{\lambda_{n'}^* - P + 1}$ clinches object $x_{k_{\lambda_{n'}^* - P + 1}}$, or agent i clinches x_i , where the latter case is only possible if $\lambda_{n'}^* = P + n' - 1$. However, if agent $k_{\lambda_{n'}^* - P + 1}$ clinches object $x_{k_{\lambda_{n'}^* - P + 1}}$, then object $x_{k_{\lambda_{n'}^* - P + 1}}$ has been offered to an active non-lurker at $h_{n'}^*$, and so $x_{k_{\lambda_{n'}^* - P + 1}}$ cannot be the next lurked object along the initial passing path (Remark 5), a contradiction. Therefore, it must be that $\lambda_{n'}^* = P + n' - 1$, and agent i is the agent that moves at $h_{n'}^*$.

Parts (b). As shown in part (a), for $n' = 1, \dots, n-1$, there are $\lambda_{n'}^* = P + n' - 1$ lurked objects at $h_{n'}^*$, which immediately implies that $h_1^* \not\subseteq h_2^* \not\subseteq \dots \not\subseteq h_{n-2}^* \not\subseteq h_{n-1}^*$ (because the number of lurked objects only grows as we go down the initial passing path).

¹⁴This is implicitly assuming that $\lambda_{n'}^* > P$. An analogous argument works for the case that $\lambda_{n'}^* \leq P$, but, for brevity, this argument is omitted.

It remains to show that $h_{n-1}^* \not\subseteq h_n^*$. By way of contradiction, assume that $h_n^* \subseteq h_{n-1}^*$. Then, $\lambda_n^* \leq \lambda_{n-1}^* = P + n - 2$, and the lurked objects at h_n^* are $\{x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{\lambda_n^*}}\}$. If a lurked object is clinched at h_n^* , then h_n^* is the terminating history, and there is no passing action at h_n^* (Lemma 14). However, this contradicts that $x_{k_{\lambda_n^*+1}}$ is the next lurked object on the initial passing path. So, it must be an unlurked object that is clinched. By the structure of $\succ_n$, it must be $k_{\lambda_n^*+1}$ that clinches $x_{k_{\lambda_n^*+1}}$. But then, $x_{k_{\lambda_n^*+1}}$ has been offered to active nonlurker at h_n^* , and so $x_{k_{\lambda_n^*+1}}$ cannot be the next lurked object along the initial passing path (Remark 5), which is a contradiction. Therefore, $h_{n-1}^* \not\subseteq h_n^*$.

Part (c). Part (a) shows this for $n' \leq n - 1$. So, we must show $\lambda_n^* = P + n - 1$. Notice that $h_{n-1}^* \not\subseteq h_n^*$ implies that $\lambda_n^* \geq \lambda_{n-1}^* = P + n - 2$, while the structure of $\succ_n$ (in particular, the tie between agent i and k_n), implies that $\lambda_n^* \leq P + n - 1$. Thus, we need to show $\lambda_n^* \neq P + n - 2$. Assume that $\lambda_n^* = P + n - 2$. Then, the lurked objects are $x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n-2}}$, and the agents coded in step 1 are $j_1, \dots, j_P, k_1, \dots, k_{n-2}, k_{n-1}$. If a lurked object is clinched at h_n^* , then this is the terminating history, which contradicts that $x_{k_{n-1}}$ is the next lurked object along the initial passing path (Lemma 14). If an unlurked object is clinched, then it must be k_{n-1} clinching $x_{k_{n-1}}$, but since this is offered to an active non-lurker, $x_{k_{n-1}}$ cannot be the next lurked object along the initial passing path (Remark 5), a contradiction. Therefore, $\lambda_n^* = P + n - 1$.

Part (d). By part (a), agent i moves at $h_{n'}^*$ in $\Gamma_{n'}$, and, since i ties in $\succ_{n'}$, object x_i is unlurked. Therefore, all lurked objects are immediately assigned to their lurkers, which delivers the result.

Part (e). To show $h_{n-1}^* \not\subseteq h_{n+1}^*$, assume not. Then, $h_{n+1}^* \subseteq h_{n-1}^*$, and $\lambda_{n+1}^* = P + \bar{n} - 1$ for some $\bar{n} \leq n - 1$. So, the lurked objects at h_{n+1}^* are $x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{\bar{n}}}$, and the agents coded in step 1 are $j_1, \dots, j_P, k_1, \dots, k_{\bar{n}}$. Since $\bar{n} \leq n - 1$, we know that $x_{k_{\bar{n}}}$ must be the next lurked object on the initial passing path. An argument analogous to those given above delivers a contradiction.

To show $\lambda_{n+1}^* \geq P + n - 1$, note that $h_{n-1}^* \not\subseteq h_{n+1}^*$ implies $\lambda_{n+1}^* \geq \lambda_{n-1}^* = P + n - 2$. Thus, we must just show that $\lambda_{n+1}^* \neq P + n - 2$. So, assume this was the case. Then, the lurked objects are $x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n-2}}$, and the agents coded in step 1 are $j_1, \dots, j_P, k_1, \dots, k_{n-2}, k_{n-1}$. If a lurked object is clinched at h_{n+1}^* , then this is the terminating history, which contradicts that $x_{k_{n-1}}$ is the next lurked object along the initial passing path (Lemma 14). If an unlurked object is clinched, then it must be k_{n-1} clinching $x_{k_{n-1}}$, but since this is offered to an active non-lurker, $x_{k_{n-1}}$ cannot be the next lurked object along the initial passing path (Remark 5), a contradiction. Therefore, $\lambda_{n+1}^* \geq P + n - 1$.

This completes the proof of Lemma 25. ■

Continuing with the proof of Lemma 20, we first show the first statement that at h_{n+1}^* in Γ_{n+1} , there is an agent $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n, i$ such that ℓ is an active non-lurker at h_{n+1}^* that does not move at h_{n+1}^* , and $x_i \in C_\ell^{\mathbb{Q}}(h_{n+1}^*)$. By Lemma 25, we have (i) $h_{n-1}^* \not\subseteq h_n^*, h_{n+1}^*$ (ii) $\lambda_n^* = P + n - 1$ and (iii) $\lambda_{n+1}^* \geq P + n - 1$. In particular, the lurked objects at h_n^* are $\{x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n-1}}\}$. Since there is a tie in $\succ_n$, there are two active non-lurker roles at h_n^* , and both of these roles have been offered to clinch x_i at h_n^* . Let s be the role that moves at h_n^* , and s' be the other active non-lurker that does not move at h_n^* .

Case 1: x_{k_n} is the next lurked object along the initial passing path of the game form. Since x_{k_n} is the next lurked object along the initial passing path, it must be i that moves at h_n^* and clinches x_i , i.e., $\sigma_n(s) = i$.¹⁵ Further, we have $h_n^* \not\subseteq h_{n+1}^*$. To see this, note that if not, then

¹⁵Agent k_n cannot move at h_n^* , because then x_{k_n} would have been offered to an active non-lurker at h_n^* , which contradicts that x_{k_n} is the next lurked object along the initial passing path. Nor can it be any $j_1, \dots, j_P, k_1, \dots, k_{n-1}$

$h_{n+1}^* \subseteq h_n^*$, and x_{k_n} is not lurked at h_{n+1}^* . Thus, it cannot be a lurked object that is clinched at h_{n+1}^* , because this would imply that h_{n+1}^* is the terminating history (Lemma 14), which contradicts that x_{k_n} becomes lurked along the initial passing path. So, the object clinched at h_{n+1}^* must be unlurked, and so the set of lurked objects is $\{x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{\bar{n}-1}}\}$, where $x_{k_{\bar{n}}}$ is the unlurked object that is clinched, and $\bar{n} \leq n$, which follows because $h_{n+1}^* \subseteq h_n^*$. But then, $x_{k_{\bar{n}}}$ is offered to an active non-lurker at h_{n+1}^* , which contradicts that it is the next lurked object along the initial passing path (Remark 5). Therefore, $h_n^* \not\subseteq h_{n+1}^*$.

Since x_{k_n} is the next lurked object along the initial passing path, we must have x_{k_n} becoming lurked at some h' such that $h_n^* \not\subseteq h' \subseteq h_{n+1}^*$. But, notice that there is still some role r such that, at h' , r is an active non-lurker, and $x_i \in C_r^{\subseteq}(h')$. Thus, x_i cannot be the next lurked object along the initial passing path. Therefore, for i to be ranked immediately after k_n in $\succ_{n+1}$, she must clinch x_i while it is unlurked, either at h_{n+1}^* , or in the immediately following clinching phase of the game.

We next claim that in Γ_{n+1} , $\sigma_{n+1}^{-1}(i) \neq s, s'$. To see this, first note that if $\sigma_{n+1}^{-1}(i) = s$, then i has the same role in Γ_n and Γ_{n+1} , and thus would once again clinch at h_n^* in Γ_{n+1} , which contradicts $h_n^* \not\subseteq h_{n+1}^*$. Therefore, $\sigma_{n+1}^{-1}(i) \neq s$. Next, assume that $\sigma_{n+1}(s') = i$. Notice that role s' cannot be the terminator role, by Lemma 15(iii) and the fact that $x_i \in C_s(h_n^*)$ and $x_i \in C_{s'}^{\subseteq}(h_n^*)$. Thus, only objects that are unlurked at h_n^* are possible for role s' , and so if $\sigma_{n+1}(s') = i$, since x_i is i 's top unlurked object, she would clinch it at some history $h' \not\subseteq h_n^* \subseteq h_{n+1}^*$, which is a contradiction. Therefore, $\sigma_{n+1}^{-1}(i) \neq s, s'$.

We showed above that s' is not the terminator role. If s is the terminator role, then, when i clinches at h_n^* , we conclude that x_i is her top possible object among all of those that are available, by Lemma 15(ii). This implies that i cannot be in a role that is a lurker at h_n^* . So, we have shown that in Γ_{n+1} , agent i is not a lurker at h_n^* , nor is she role s or s' . Thus, i is not active at h_n^* in Γ_{n+1} , and so there must be some agent $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n$ such that $\sigma_{n+1}^{-1}(\ell) = s$ or s' . But then, since i is unlurked at h_{n+1}^* , we have that $x_i \in C_{\ell}^{\subseteq}(h_{n+1}^*)$, as desired.

If s is not the terminator role, we once again claim that i cannot be in a role that is a lurker at h_n^* . Indeed, if this were true, then some agent j who is receiving a lurked object is not a lurker at h_n^* . Therefore, this agent must be in the terminator role, and clinch at h_{n+1}^* . Since the terminator role is not s or s' , it is not yet active at h_n^* , and so j is not active at h_n^* in Γ_{n+1} . Therefore, there must be some $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n$ such that $\sigma_{n+1}^{-1}(\ell) = s$ or s' , and that is still active when j clinches at h_{n+1}^* , which implies that $x_i \in C_{\ell}^{\subseteq}(h_{n+1}^*)$, as desired.

Case 2: x_{k_n} is not the next lurked object along the initial passing path. By Lemma 25, at h_n^* , there are $P + n - 1$ lurked objects. This implies that both i and k_n are coded in step 1 of the coding algorithm for Γ_n , and thus that the first unlurked object that is clinched is either x_i or x_{k_n} .¹⁶ This gives rise to two subcases.

Case 2.1: x_{k_n} is the first unlurked object that is clinched in the coding algorithm in Γ_n . In this case, $\sigma_n(s') = i$, and there is some history $\tilde{h} \not\subseteq h_n^*$ such that $x_{k_n} \in C_i(\tilde{h})$.

CLAIM 1: *The following are true: (a) $h_{n-1}^* \not\subseteq h_{n+1}^* \not\subseteq h_n^*$ and (b) agent k_n clinches x_{k_n} at h_{n+1}^* in Γ_{n+1} , and x_{k_n} is unlurked at this history.*

Proof of Claim 1. Part (a). First notice that $h_{n-1}^* \not\subseteq h_{n+1}^*$ follows from Lemma 25. So, we must show that $h_{n+1}^* \not\subseteq h_n^*$. Towards a contradiction assume that $h_n^* \subseteq h_{n+1}^*$. Since $h_{n-1}^* \not\subseteq h_n^*$, we have $h_{n-1}^* \not\subseteq h_n^* \subseteq h_{n+1}^*$. Lemma 25 also implies that $\lambda_n^* = P + n - 1$. Since i does not move

, because then they would be clinching a lurked object, and so h_n^* is the terminating history, which again contradicts that x_{k_n} is the next lurked object along the initial passing path.

¹⁶Note that this does not necessarily mean that the object clinched at h_n^* is x_i or x_{k_n} .

at h_n^* in Γ_n , it must be some $j_1, \dots, j_P, k_1, \dots, k_n$ that does. If a lurked object is clinched at h_n^* , then h_n^* is the terminating history. It also implies that agent k_n is a lurker for some lurked object, and therefore in step 1 of the coding algorithm, some agent takes the object k_n lurks, and he ends the step by clinching x_{k_n} , which is unlurked. This means that x_{k_n} is his favorite object that is unlurked at h_n^* . Now, consider Γ_{n+1} , and note that $h_n^* \subseteq h_{n+1}^*$ and h_n^* being the terminating history implies that $h_n^* = h_{n+1}^*$. In Γ_{n+1} , the set of lurked objects is the same as in Γ_n , so x_{k_n} is again the first unlurked object that is clinched in step 1 of the coding algorithm. But, since $h_n^* = h_{n+1}^*$, there is again an agent in role s' who is an active non-lurker at h_{n+1}^* , and so this agent would once again tie with k_n in $\succ_{n+1}$, a contradiction. Therefore, it must be that k_n is the agent that moves at h_n^* in Γ_n , which means that x_{k_n} has been offered to both active non-lurker roles at h_n^* . Since we assumed that $h_n^* \subseteq h_{n+1}^*$, it is impossible for k_n to be ranked $(P+n)^{th}$ strictly, without ties, in $\succ_{n+1}$,¹⁷ which is a contradiction. Thus, we have shown that $h_{n+1}^* \not\subseteq h_n^*$, which is part (a).

Part (b). Part (a) plus Lemma 25 implies that $\lambda_{n+1}^* = P+n-1$. Additionally, $h_{n+1}^* \not\subseteq h_n^*$ means that h_{n+1}^* is not the terminating history, so it must be an unlurked object that is clinched there. Thus, since k_n is ordered $(P+n)^{th}$ without ties, it must be that k_n clinches x_{k_n} at h_{n+1}^* , and x_{k_n} is unlurked. ■

By Lemma 25, the agent that moves at h_{n-1}^* must be agent i , and therefore, at h_{n-1}^* , there are two active non-lurker roles that both have been offered x_i . Let the role that moves at h_{n-1}^* be denoted r , and the other active non-lurker at h_{n-1}^* be denoted r' . Thus, by definition, $\sigma_{n-1}(r) = i$.

We claim that in Γ_{n+1} , i cannot be active at h_{n-1}^* . At h_{n-1}^* , there are $P+n-2$ active lurker roles, and two active non-lurker roles, r and r' . First, it is clear that $\sigma_{n+1}(r) \neq i$, because otherwise i is in the same role in Γ_{n-1} and Γ_{n+1} , and so would clinch at h_{n-1}^* in Γ_{n+1} , which contradicts $h_{n-1}^* \not\subseteq h_{n+1}^*$ from Claim 1. Second, assume that in Γ_{n+1} , agent i is in a lurker role for a lurked object at h_{n-1}^* , say y . By part (b) of Claim 1, agent k_n clinches an unlurked object at h_{n+1}^* , and so all lurkers are immediately assigned to their lurked objects, which means that i would receive y which is a contradiction.

It remains to rule out that $\sigma_{n+1}^{-1}(i) = r'$. By construction, $x_i \in C_r(h_{n-1}^*)$, where $x_i \in C_{r'}(\tilde{h})$ for some $\tilde{h} \not\subseteq h_{n-1}^*$. This implies that role r' cannot be the terminator role, by Lemma 15(iii), and the fact that $x_i \in C_r(h_{n-1}^*)$. Since role r' is not the terminator role, only unlurked objects are possible for role r' , by Lemma 15(iv). As x_i is agent i 's most preferred unlurked object, by greedy strategies, she would clinch at $\tilde{h}$, which is a contradiction. Therefore, i is not active at h_{n-1}^* in Γ_{n+1} .

We also claim that i is not active at h_{n-1}^* in Γ_n , either. The arguments are the same as above for Γ_{n+1} , except for the case in which i lurks some lurked object at h_{n-1}^* . This is ruled out by the fact that $\sigma_n(s') = i$, and s' is a non-lurker at h_{n-1}^* .

Next, we claim that $\sigma_{n+1}(s) \neq i$. To see this, recall that $\sigma_n(s') = i$, and, as we showed, i is not active at h_{n-1}^* in Γ_n or Γ_{n+1} . This means that $s' \neq r, r'$, or in other words, s' is a role that becomes active after h_{n-1}^* . Thus, we must have $s = r$ or r' , and so role s is active at h_{n-1}^* , which implies that $\sigma_{n+1}(s) \neq i$.

Next, we claim that $\sigma_{n+1}(s') = k_n$. Indeed, since $h_{n-1}^* \not\subseteq h_{n+1}^* \not\subseteq h_n^*$ and k_n moves at h_{n+1}^* , k_n must be in role either s or s' . If $\sigma_{n+1}(s) = k_n$, then, since she does not tie in $\succ_{n+1}$, she must clinch x_{k_n} at some history h' such that $h_{n-1}^* \not\subseteq h' \not\subseteq \hat{h}$, where $\hat{h}$ is the history at which role s'

¹⁷Note that x_{k_n} cannot be the next lurked object, so, there must be no newly lurked objects at h_{n+1}^* (Remark 5). If k_n clinches at h_{n+1}^* , she would tie with the other active non-lurker. If some other agent clinches at h_{n+1}^* , then either this agent is ranked strictly ahead of k_n , or she ties with k_n , which again is a contradiction.

is offered to clinch x_{k_n} . This implies that $\sigma_n(s) \neq k_n$, as else in Γ_n , she would also clinch at h' . So, in Γ_n , $\sigma_n(s) = k_{n'}$ for some $n' < n$ or $\sigma_n(s) = j_p$ for some $p = 1, \dots, P$. This implies that h_n^* is the terminating history. Furthermore, k_n is in the lurker role for some object $x' \in \{x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n-1}}\}$, which implies that k_n strictly prefers x' to x_{k_n} . But then, since $\sigma_{n+1}(s) = k_n$, agent k_n is in the terminator role in Γ_{n+1} , and thus x' is a possible outcome for her, she would not choose to clinch x_{k_n} first at h_{n+1}^* , a contradiction. Therefore, $\sigma_{n+1}(s') = k_n$.

Concluding the argument for Case 2.1, because k_n clinches an unlurked object at h_{n+1}^* in Γ_{n+1} , all agents $j_1, \dots, j_P, k_1, \dots, k_{n-1}$ must be in the lurker role for their respective objects. Therefore, none of them are in role s . As just shown, $\sigma_{n+1}(s) \neq k_n$ or i , either. All of this means that $\sigma_{n+1}(s) = \ell$ for some $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n, i$, and in Γ_{n+1} , we have $x_i \in C_\ell^\subseteq(h_{n+1}^*)$, as desired.

Case 2.2: x_i is the first unlurked object that is clinched in step 1 of the coding algorithm in Γ_n . In this case, we have that $\sigma_n(s') = k_n$, and $x_i \in C_{s'}(\tilde{h})$ for some $\tilde{h} \not\subseteq h_n^*$. There are two further subcases:

Case 2.2.1: $\sigma_n(s) \neq i$. In this subcase, $\sigma_n(s)$ is one of $j_1, \dots, j_P, k_1, \dots, k_{n-1}$, and is clinching a lurked object at h_n^* . This implies that h_n^* is the terminating history, and s is the terminator role, which also means that we have $h_{n-1}^* \not\subseteq h_{n+1}^* \subseteq h_n^*$. This combined with Lemma 25 implies that there are $P + n - 1$ lurkers at h_n^* , and the structure of $\succ_{n+1}$ means that x_{k_n} is the first unlurked object clinched in step 1 of Γ_{n+1} , and, at h_{n+1}^* , x_{k_n} has not been offered to the active non-lurker who does not move at h_{n+1}^* .

We also claim that role s cannot be active at history h_{n-1}^* . Indeed, since i clinches at h_{n-1}^* in Γ_{n-1} and ties, we know that there are two active non-lurker roles, say r and r' , and they both have been offered x_i . If role s were one of these roles, then, since s is the terminator role, Lemma 15 implies that $x_i \notin C_{s'}(\tilde{h})$, which is a contradiction. This implies that role s is a role that becomes active after h_{n-1}^* . Since there is only one new lurker between h_{n-1}^* and h_{n+1}^* , this further implies that role s' must have been active at h_{n-1}^* , and $x_i \in C_{s'}^\subseteq(h_{n-1}^*)$.

We next claim that $\sigma_{n+1}(s') \neq i$. To see why this is true, notice that s' is not the terminator role (because that is role s). Thus, only unlurked objects are possible for role s' (Lemma 15(iv)), and, since we know that x_i is i 's favorite unlurked object, if she were in role s' , she would clinch at $\tilde{h} \not\subseteq h_{n+1}^*$, a contradiction. Therefore, $\sigma_{n+1}(s') \neq i$.

Now, if it is one of the $j_1, \dots, j_P, k_1, \dots, k_{n-1}$ that moves at h_{n+1}^* , then h_{n+1}^* is the terminating history, and so $h_{n+1}^* = h_n^*$. This implies that x_i has been offered to the agent in role $\sigma_{n+1}(s')$ (who is not coded in step 1). As we just showed that $\sigma_{n+1}(s') \neq i$, we have $\sigma_{n+1}(s') = \ell$ for some $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n$, and $x_i \in C_\ell(h_{n+1}^*)$ in Γ_{n+1} , as desired.

Concluding subcase 2.2.1, assume that it is k_n that moves at h_{n+1}^* in Γ_{n+1} . This means that k_n is in role s or s' in Γ_{n+1} . Note that we cannot have $\sigma_{n+1}(s') = k_n$, because if this were true, then k_n has the same role in Γ_n as in Γ_{n+1} , and would pass at all histories in Γ_{n+1} , just as she did in Γ_n . Therefore, $\sigma_{n+1}(s) = k_n$. Again, as we know that $\sigma_{n+1}(s') \neq i$, we have that $\sigma_{n+1}(s') = \ell$ for some $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n$, and $x_i \in C_\ell(h_{n+1}^*)$ in Γ_{n+1} , as desired.

Case 2.2.2: $\sigma_n(s) = i$. In this subcase, i clinches x_i at h_n^* . If $h_n^* \subseteq h_{n+1}^*$, then notice that at h_n^* in Γ_{n+1} , there are two active non-lurker roles, s and s' , that have been offered x_i . We claim that $\sigma_{n+1}^{-1}(i) \neq s, s'$. First, it is clear that $\sigma_{n+1}(s) \neq i$, as otherwise, i would clinch at h_n^* in Γ_{n+1} , just as she did in Γ_n . To see that $\sigma_{n+1}(s') \neq i$, notice that role s' cannot be the terminator role, by Lemma 15 and the fact that $x_i \in C_s(h_n^*)$ and $x_i \in C_{s'}^\subseteq(h_n^*)$. Thus, only unlurked objects are possible for role s' , and so if $\sigma_{n+1}(s') = i$, since x_i is i 's top unlurked object, she would clinch it at some history $h' \not\subseteq h_n^* \subseteq h_{n+1}^*$, which is a contradiction. Therefore, $\sigma_{n+1}^{-1}(i) \neq s, s'$, and so there must be some $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n$ such that $x_i \in C_\ell(h_{n+1}^*)$, as desired.

It remains to consider $h_{n+1}^* \not\subseteq h_n^*$. Then, there are $P + n - 1$ lurkers at h_{n+1}^* , and, since h_{n+1}^* is not the terminating history, it must be agent k_n that moves at h_{n+1}^* . This also implies that k_n

is in role s or s' . If $\sigma_{n+1}(s') = k_n$, then k_n is in the same role in Γ_{n+1} as in Γ_n , and would pass at h_{n+1}^* in Γ_{n+1} as she did in Γ_n , which is a contradiction. Therefore, $\sigma_{n+1}(s) = k_n$.

We claim that role s is not active at history h_{n-1}^* . Indeed, notice that because i clinches at h_{n-1}^* in Γ_{n-1} , we have that $x_i \in C_s^\subseteq(h_{n-1}^*)$. This implies that role s is not the terminator role, which follows by Lemma 15 and the fact that $x_i \in C_{s'}(h')$ for some $h' \supsetneq h_{n-1}^*$. This implies that only unlurked objects are possible for role s when she is called to play. Thus, if role s were an active non-lurker at history h_{n-1}^* , then, in Γ_n , when $\sigma_n(s) = i$, agent i is offered to clinch x_i at some $h' \subseteq h_{n-1}^*$. Since we know that only unlurked objects are possible, and x_i is i 's top unlurked object, she would clinch at $h' \subsetneq h_n^*$ in Γ_n , which is a contradiction. Since role s is not active at h_{n-1}^* , there are two roles that are not s that are active non-lurkers at h_{n-1}^* and such that both have been offered to clinch x_i . At h_{n+1}^* in Γ_{n+1} , at least one of these roles must still be active and not assigned to any agent $j_1, \dots, j_P, k_1, \dots, k_n, i$. Thus, there must be some $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n, i$ such that ℓ is an active non-lurker that does not move at h_{n+1}^* and $x_i \in C_\ell^\subseteq(h_{n+1}^*)$, as desired. This concludes the analysis of subcase 2.2.2, and hence of case 2.2.

The above shows that in all cases, there is some $\ell \neq j_1, \dots, j_P, k_1, \dots, k_n, i$ such that ℓ is an active non-lurker that does not move at h_{n+1}^* and $x_i \in C_\ell^\subseteq(h_{n+1}^*)$ in game Γ_{n+1} . Recall that, by Lemma 25, $\lambda_{n+1}^* \geq P + n - 1$. If $\lambda_{n+1}^* > P + n - 1$, then there are at least $P + n$ lurked objects at h_{n+1}^* , and the only way i can be ranked in the $(P + n + 1)^{th}$ position in $\succ_{n+1}$ is if she is coded in the first step. Since there is some agent $\ell \neq i$ such that $x_i \in C_\ell^\subseteq(h_{n+1}^*)$, i can at best tie with this agent. If $\lambda_{n+1}^* = P + n - 1$, then by the structure of $\succ_{n+1}$, it must be agent k_n that clinches at h_{n+1}^* , and there is no tie at the end of step 1. This means that ℓ is not coded in step 1, and so at $h_\emptyset^2$ (the beginning of step 2 of the coding algorithm) agent ℓ moves and is offered to clinch x_i . Now, for i to be ranked immediately after k_n , she must be ordered first in step 2 of the coding algorithm, and for i to be ordered first without ties, either she must lurk x_i and it is the first lurked object, or i must clinch x_i while there are no lurked objects and before x_i has not been offered to another active non-lurker. However, neither of these can occur because at $h_\emptyset^2$ agent ℓ was offered to clinch x_i . Therefore, in $\succ_{n+1}$, i must tie with some agent that we label k_{n+1} . This completes the proof of Lemma 20. ■

Proof of Lemma 21

We start with the following lemma. Let h_A^* denote the history at which the first clinching action is taken in Γ_A , h_B^* denotes the history at which the first clinching action is taken in Γ_B .

LEMMA 26: *Consider two games Γ_A and Γ_B , with corresponding role assignment functions σ_A and σ_B , and resulting codings $\succ_A$ and $\succ_B$. Assume that $\succ_A$ begins as $\{i, j\} \succ_A \dots$, and $\succ_B$ begins as: $j \succ_B i \dots$. Further, assume that in game Γ_A , there is some history h where j moves such that: (i) $h \subseteq h_A^*$, (ii) $x_i \in C_j(h)$ (iii) $x_j \notin C_j^\subseteq(h)$ (iv) $x_i, x_j \notin C_i^\subseteq(h)$. Then:*

(a) *If agent j clinches at h_A^* in Γ_A , then in Γ_B , agent j clinches at $h_B^* \subsetneq h_A^*$, and there is some agent $k \neq i, j$ that is an active non-lurker at h_B^* such that $x_i \in C_k(h_B^*)$.*

(b) *In $\succ_B$, agent i must tie with some other agent k .*

Proof of Lemma 26. Notice that by the structure of $\succ_A$, at history h_A^* , there are two active roles, and both are nonlurkers at h_A^* ; label the roles s and s' , and, wlog, let $\sigma_A(s) = i$ and $\sigma_A(s') = j$. Using these definitions, we can write the presumptions of the lemma as (ii) $x_i \in C_{s'}(h)$ (iii) $x_j \notin C_{s'}^\subseteq(h)$ and (iv) $x_i, x_j \notin C_s^\subseteq(h)$. Also, notice that $h \subseteq h_A^*$ implies that there are no lurkers at h , and so the only roles that may possibly be active at h are s and s' . Finally, since x_i and x_j tie for the top ranking in $\succ_A$, it must be that x_i is i 's favorite object among all objects

and x_j is j 's favorite object among all objects. Therefore, by greedy strategies, if at any history i is able to clinch x_i , she will do so, and the same for j and x_j .

Part (a). The structure of $\succ_A$ implies that $x_j \in C_s(h')$ for some $h' \not\subseteq h_A^*$. Now, consider Γ_B . The only way for j to be ranked first without ties is that $\sigma_B(s) = j$, and j clinches at $h_B^* \not\subseteq h_A^*$.¹⁸ Let $k := \sigma_B(s')$, and notice that, by the assumptions of the lemma, $x_j \notin C_s^{\subseteq}(h)$, and so $h \not\subseteq h_B^*$, and therefore $x_i \in C_{s'}^{\subseteq}(h_B^*)$. It is clear that $k \neq j$. Further, $k \neq i$ because if $k = i$, then $x_i \in C_i(h)$ in Γ_B , and thus, i would clinch x_i at $h \not\subseteq h_B^*$ in Γ_B , which contradicts that the first clinching in Γ_B is j clinching at h_B^* . Therefore, $\sigma_B(s') = k$ for some $k \neq i, j$, and k is an active non-lurker that does not move at h_B^* such that $x_i \in C_k(h_B^*)$ in Γ_B .

Part (b). If j clinches at h_A^* , then by part (a), in Γ_B , there is an agent k such that $x_i \in C_k^{\subseteq}(h_B^*)$ and k is not coded in the coding step initiated at h_B^* . Let $h_B^{**} \not\supseteq h_B^*$ be the history at which the next clinching occurs in Γ_B . Since k was offered x_i in the previous coding step, but is still active, at h_B^{**} , agent k is offered to clinch x_i again (see Remark 3, in particular footnote 49 of the main text). Thus, x_i cannot be the first lurked object on the initial passing path of the continuation game form (Remark 5), and so since i is ranked immediately following j in $\succ_B$, there must be no lurked objects at h_B^{**} . For i to be coded next, she must be active at h_B^{**} , and since there are no lurked objects, there are two active agents, i and k . If k clinches at h_B^{**} , it is obvious that i can at best tie; if i clinches at h_B^{**} , i once again ties with k , because $x_i \in C_k^{\subseteq}(h_B^{**})$.

The other possibility is that i clinches at h_A^* in Γ_A , which implies that $x_i \in C_{s'}(h')$ for some $h' \not\subseteq h_A^*$. For j to be ranked first without ties in $\succ_B$, at h_B^* in Γ_B , either (a) there are lurkers, and x_j is the first lurked object or (b) there are no lurkers, j clinches x_j , and x_j has not been offered to another non-lurker that is active at h_B^* . There are 3 cases:

Case: $\sigma_B(s') = i$. In this case, i would clinch x_i at h and would be ranked first in $\succ_B$, which is a contradiction.¹⁹

Case: $\sigma_B(s') = j$. Here, j is in the same role in both games, and therefore $\sigma_B(s) = \ell \neq i$, which follows because if $\ell = i$, then both j and i are in the same roles, and we would get the same initial orderings for $\succ_A$ and $\succ_B$, a contradiction.²⁰ This implies that $h_B^* \not\supseteq h_A^*$, because if $h_B^* \subseteq h_A^*$, then, since j is in the same role, she would clinch at h_B^* in Γ_A , a contradiction.²¹ Recall that i is ranked immediately after j in $\succ_B$, and because $x_i \in C_{\ell}^{\subseteq}(h_B^*)$, this is only possible if i ties with ℓ . To see this, assume that i was ranked strictly, i.e., $j \succ_B i \succ_B \dots$. This can only happen if either: (i) x_j is the oldest lurked object and x_i is second-oldest lurked object at h_B^* (ii) the only lurked object at h_B^* is x_j , while x_i is unlurked at h_B^* , i clinches x_i at h_B^* , and x_i has not been previously offered to another active non-lurker or (iii) agent j is the only agent coded in step 1 of Γ_B , and agent i is the first agent coded in step 2, and she is ranked strictly. But, as x_i has been offered to both j and ℓ weakly prior to h_A^* and $h_A^* \not\subseteq h_B^*$, none of these three possibilities can hold, and therefore, i must tie in $\succ_B$.²²

¹⁸The only other way for j to be ranked first without ties is that x_j is the first lurked object; however, this cannot obtain, because $x_j \in C_s(h')$ at some history h' where there are no lurkers.

¹⁹Note that x_j has not been offered to any agent at h , by the presumptions of the lemma.

²⁰Notice that in Γ_A , at h_A^* , only two agents are active, i and j . If these agents are in the same roles in Γ_B , then $\succ_B$ would also begin $\{i, j\} \succ_B \dots$.

²¹The case $h_B^* = h_A^*$ is ruled out because i moves at h_A^* in Γ_A , and this history is controlled by role s , not s' .

²²Because x_i has been offered to both j and ℓ weakly prior to h_A^* , it cannot be the first or second lurked object (see Remark 5), which rules out (i). Since $x_i \in C_{\ell}^{\subseteq}(h_B^*)$, (ii) cannot hold. For (iii), since j is the only agent coded in step 1, but $x_i \in C_{\ell}^{\subseteq}(h_B^*)$, we have that ℓ begins the round 2 continuation game by being offered x_i (plus possibly other objects) at h_B^{**} . Since $x_i \in C_{\ell}(h_B^{**})$, x_i cannot be the first lurked object in round 2 (again by Remark 5), nor can i be ranked first in this round and strictly so (for i to be ranked strictly first in the continuation game, either x_i —which Lemma 3 tells us i gets—has to be the first lurked object, or there are no lurked objects and i clinches x_i while it is unlurked and it has not been offered to the other active non-lurker; neither of these can happen because the continuation game begins with x_i being offered to ℓ).

Case: $\sigma_B(s') = \ell'$ for some $\ell' \neq i, j$. First, notice that $\sigma_B(s) = \ell$ for some $\ell \neq i$. To see this, assume that $\ell = i$. Then, i is in the same role in Γ_A and Γ_B . This implies that $h_B^* \not\subseteq h_A^*$, because if h_A^* is reached in Γ_B , i would clinch there, and be ranked above j . But, $h_B^* \not\subseteq h_A^*$ implies that j is not ranked first in $\succ_B$ (since she is not yet active at h_B^*), which is a contradiction.

If $\sigma_B(s) = j$, then for j to be ranked first in $\succ_B$, either (a) x_j is the first lurked object on the path to h_B^* or (b) there are no lurked objects at h_B^* , j clinches x_j at h_B^* , and x_j has not been offered to another active non-lurker. Notice that $h_B^* \not\supseteq h$,²³ which implies that $x_i \in C_{\ell'}^c(h_B^*)$. But, since x_i has been offered to agent ℓ' , it is impossible for i to be ranked immediately after j in $\succ_B$ without ties. Therefore, i must tie in $\succ_B$.

If $\sigma_B(s) \neq j$, then roles s and s' are assigned to agents ℓ and ℓ' in Γ_B , neither of which are j or i . So, for j to be ranked first without ties, x_j must be the first lurked object (and be lurked by either ℓ or ℓ'), and j must clinch it at some $h_B^* \not\supseteq h_A^*$. For i to be ranked second without ties in this case, there must be two lurked objects at h_B^* ,²⁴ and x_i must be the second lurked object (after x_j). But, at the history $h'' \not\supseteq h_A^*$ where x_j becomes lurked, one of agents ℓ or ℓ' is an active non-lurker who has been previously offered to clinch x_i , and so x_i cannot be the next lurked object, a contradiction. This completes the proof of Lemma 26. ■

Continuing with the proof of Lemma 21, first, consider $Q = 1$. Then, all agents $j_1, \dots, j_P, k_1, \dots, k_n, i$ are coded in step 1 of game Γ_n . By Remark 6, $x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n-1}}$ all become lurked on the initial passing path of the game form, and further, since $n \geq 2$, we can apply Lemma 20, which completes the proof for this case.

It remains to consider $Q \geq 2$. Since at least $P+1$ agents are coded in step 1 (see the discussion at the beginning of the proof of Lemma 5), all agents j_p have been coded in the first step, and so the agent who is coded first in step Q of the coding algorithm of Γ_n is $k_{\bar{n}}$ for some $\bar{n} < n$. So, the subcoding of $\succ_n$ starting from step Q is:

$$k_{\bar{n}} \succ_n k_{\bar{n}+1} \succ_n \dots \succ_n k_{n-1} \succ_n \{i, k_n\}.$$

Consider the sequence of games $\Gamma_{\bar{n}}, \Gamma_{\bar{n}+1}, \dots, \Gamma_n, \Gamma_{n+1}$. Notice that the codings for all of these games are exactly the same, up to agent $k_{\bar{n}-1}$. Therefore, by Lemma 4, all agents $j_1, \dots, j_P, k_1, \dots, k_{\bar{n}-1}$ are in the same roles in all of these games. In particular, agent $k_{\bar{n}-1}$ is the last agent coded in step $Q-1$ in all of these games, and the initiating history of the continuation game that begins step Q is the also the same in all of these games; label this history $h_{\emptyset}^Q$. Now, applying the coding algorithm to the sequence of continuation games of $\Gamma_{\bar{n}}, \dots, \Gamma_n, \Gamma_{n+1}$ starting from history $h_{\emptyset}^Q$, we get the sub-codings:

$$\begin{aligned} & \{i, k_{\bar{n}}\} \succ_{\bar{n}} \dots \\ & k_{\bar{n}} \succ_{\bar{n}+1} \{i, k_{\bar{n}+1}\} \succ_{\bar{n}+1} \dots \\ & \vdots \\ & k_{\bar{n}} \succ_n k_{\bar{n}+1} \succ_n \dots \succ_n k_{n-1} \succ_n \{i, k_n\} \succ_n \dots \\ & k_{\bar{n}} \succ_{n+1} k_{\bar{n}+1} \succ_{n+1} \dots \succ_{n+1} k_n \succ_{n+1} i \dots \end{aligned}$$

There are two cases.

²³In case (a), this follows because there are no lurkers at h ; in case (b), it follows from the assumption of the lemma that $x_j \notin C_s^c(h)$.

²⁴Since j clinches at h_B^* , if there is no other lurked object at h_B^* , the only active agents are ℓ, ℓ' , and j , and so one of ℓ or ℓ' will be ranked above i in $\succ_B$, which is a contradiction.

Case 1: $\bar{n} < n$. In this case, we can apply Lemma 20 to the game form starting from $h_\emptyset^Q$ to conclude that i must tie in $\succ_{n+1}$. To see this, simply note that upon reindexing to start from $h_\emptyset^Q$ rather than $h_\emptyset$, the condition “ $n \geq 2$ ” becomes “ $n \geq \bar{n} + 1$ ”. Then, we have that $x_{k_{\bar{n}}}, \dots, x_{k_{n-1}}$ all become lurked on the initial passing path of the game form starting from $h_\emptyset^Q$, which follows from Remark 6, $n \geq \bar{n} + 1$, and the fact that i is coded in the initial step of the continuation game of Γ_n starting from $h_\emptyset^Q$. Thus, all of the conditions of Lemma 20 are satisfied.

Case 2: $\bar{n} = n$. In this case, the games we are concerned with are Γ_n and Γ_{n+1} , with subcodings:

$$\begin{aligned} \{i, k_n\} &\succ_n \dots & (A) \\ k_n &\succ_{n+1} i \dots \end{aligned}$$

Notice that here, we can no longer apply Lemma 20, since we do not have at least two games in which i ties in the sequence. Our goal is to apply Lemma 26 instead, but to do so, we must show that the conditions (i)-(iv) of Lemma 26 are satisfied at $h_\emptyset^Q$.

For each coding step $q = 1, \dots, Q$ of game Γ_n , let h_n^{q*} denote history at which the first object is clinched in the q^{th} coding step, and let $h_n^{\emptyset q}$ denote the initial history that begins the continuation game for the next step, after all of the agents in step $q - 1$ are coded (in particular, $h_n^{\emptyset 1} = h_\emptyset$, and $h_n^{1*} = h_n^*$ in our earlier notation). In $\succ_n$, all agents who are coded in steps $q < Q$ are ranked strictly, without ties. Let k_{nq} denote the agent who is coded *last* in the q^{th} step. With this notation, the subcoding from the q^{th} step is:

$$k_{nq-1+1} \succ_n k_{nq-1+1} \succ_n \dots \succ_n k_{nq},$$

where we define $n^0 = 0$. It is possible that $k_{nq-1+1} = k_{nq}$, in which case only one agent is coded in step q . Since there are no ties, agent k_{nq} ends the coding step by clinching an unlurked object that has not been offered to another non-lurker who is active at h_n^{q*} .

CLAIM 1: *Assume that $Q > 1$, and that at h_n^{1*} , there is an agent ℓ that is an active non-lurker at h_n^{1*} that does not move at h_n^{1*} and $x_i \in C_\ell^\Xi(h_n^{1*})$. For all $q < Q$, there is an agent $\ell \neq k_1, \dots, k_{nq}, i$ such that ℓ is an active nonlurker at h_n^{q*} that does not move at h_n^{q*} , and $x_i \in C_\ell^\Xi(h_n^{q*})$.*

Claim 1 (whose proof can be found starting with the next paragraph) implies that when we reach step Q in Γ_n , at the initiating history of the continuation game $h_n^{\emptyset Q}$ that begins this step, there is some agent $\ell \neq k_1, \dots, k_{n-1}, i$ such that $x_i \in C_\ell(h_n^{\emptyset Q})$. Since the subcodings for $\succ_n$ in this step begin with a tie between i and k_n (see Equation A), it must be that $\ell = k_n$. Finally, we apply Lemma 26 by setting $A = n$, $B = n + 1$, $h = h_n^{\emptyset Q}$, $j = k_n$, and $i = i$ to conclude that i must tie in $\succ_{n+1}$.²⁵ To complete the proof of Lemma 21, we hence only need to prove Claim 1.

Proof of Claim 1 (stated above). By supposition, there is an agent ℓ that is an active nonlurker at h_n^{1*} that does not move at h_n^{1*} , and $x_i \in C_\ell^\Xi(h_n^{1*})$. It is clear that ℓ is not coded in step 1 (since there is no tie in step 1), and so $\ell \neq k_1, \dots, k_{n1}$. To see that $\ell \neq i$, note that if $\ell = i$, then step 2 begins with agent i being offered to clinch x_i . If i is not coded in step 2, then step 3 begins with

²⁵Condition (i) of Lemma 26 is immediate. Condition (ii) was just shown. Condition (iii) holds because, if $x_{k_n} \in C_{k_n}(h_n^{\emptyset Q})$, then k_n would immediately clinch it at $h_n^{\emptyset n}$, and would not tie with i in $\succ_n$. Condition (iv) is also immediate, as i has not yet been called to move at $h_n^{\emptyset Q}$.

i being offered x_i , etc. The same continues up to and including step Q , in which i is coded. Since i is coded first in step Q (tying with k_n) x_i is her top object among those that remain at the beginning of step Q . Since $x_i \in C_i(h_n^{(Q-1)*})$, agent i begins step Q by being offered to clinch x_i at the initial history of this step. Since x_i is her top remaining object, she would clinch it, and thus would not tie with k_n , which is a contradiction. Thus, the statement holds for $q = 1$.

Now, consider step $q = 2$ of game Γ_n , which begins at $h_n^{\emptyset_2}$ and produces the subcoding:

$$k_{n^{1+1}} \succ_n k_{n^{1+2}} \succ_n \cdots \succ_n k_{n^2} \succ_n \cdot$$

Case 1: $n^2 = n^1 + 1$. Then only one agent, agent $k_{n^{1+1}}$, is coded in step 2 of game Γ_n , which begins with the continuation game that starts at history $h_n^{\emptyset_2}$. The result from step 1 implies that at $h_n^{\emptyset_2}$, some agent $\ell \neq k_1, \dots, k_{n^1}, i$ moves and $x_i \in C_\ell(h_n^{\emptyset_2})$.

Since $k_{n^{1+1}}$ is the only agent coded in step 2 of Γ_n , and does not tie, she must clinch $x_{k_{n^{1+1}}}$ at h_n^{2*} in Γ_n while it is unlurked, and before it is offered to another active non-lurker. Now, since $\succ_n$ and $\succ_{n^{1+1}}$ are the same up til agent k_{n^1} , Lemma 4 implies that $h_n^{\emptyset_2} = h_{n^{1+1}}^{\emptyset_2}$; for shorthand, define $h^{\emptyset_2} := h_n^{\emptyset_2} = h_{n^{1+1}}^{\emptyset_2}$. The second step continuation games of $\Gamma_{n^{1+1}}$ and Γ_n both start from $h^{\emptyset_2}$, and lead to the initial subcodings:

$$\{i, k_{n^{1+1}}\} \succ_{n^{1+1}} \cdots$$

$$k_{n^{1+1}} \succ_n \cdots$$

Let role s be the role that moves at $h^{\emptyset_2}$, and role s' be the second role that becomes active on the initial passing path of the game form starting from $h^{\emptyset_2}$. These two roles exist because there is an initial tie in $\succ_{n^{1+1}}$, and in $\Gamma_{n^{1+1}}$, s and s' are assigned to $k_{n^{1+1}}$ and i , in some manner. If $\sigma_{n^{1+1}}(s) = i$, then i would clinch at $h^{\emptyset_2}$ in $\Gamma_{n^{1+1}}$, and would not tie, a contradiction. Therefore, $\sigma_{n^{1+1}}(s) = k_{n^{1+1}}$, which implies that $x_{k_{n^{1+1}}} \notin C_{k_{n^{1+1}}}(h_n^{\emptyset_2})$; indeed, if this were true, then $k_{n^{1+1}}$ would clinch it at $h_{n^{1+1}}^{\emptyset_2}$ in $\Gamma_{n^{1+1}}$, which contradicts that $k_{n^{1+1}}$ ties in $\succ_{n^{1+1}}$.

Now, if $\sigma_n(s) = k_{n^{1+1}}$, then $k_{n^{1+1}}$ is in the same role in both games, and so it must be i that clinches at $h_{n^{1+1}}^{2*}$, which means that $x_i \in C_{s'}(h_{n^{1+1}}^{2*})$.²⁶ It also means that $h_n^{2*} \not\supseteq h_{n^{1+1}}^{2*}$, and that $\sigma_n(s') \neq i$, and so, there exists some agent $\ell' \neq i$ such that in Γ_n , $x_i \in C_{\ell'}(h_n^{2*})$, which is what we wanted to show.

Last, if $\sigma_n(s) \neq k_{n^{1+1}}$, then $\sigma_n(s') = k_{n^{1+1}}$. Thus, in this case, there is some agent other agent ℓ such that $\sigma_n(s) = \ell$. Again, $\ell \neq i$, because $x_i \in C_s(h_n^{\emptyset_2})$. Thus, when $k_{n^{1+1}}$ clinches at h_n^{2*} in Γ_n , we have $x_i \in C_\ell^{\neq}(h_n^{2*})$, as desired.

Case 2: $n^2 > n^1 + 1$. Consider games $\Gamma_{n^{1+1}}, \Gamma_{n^{1+2}}, \dots, \Gamma_n$ and notice that the codings for all of these games are equivalent up to agent k_{n^1} . Therefore, by Lemma 4, all agents $k_1, \dots, k_{n^1}$ are in the same roles in all of these games, and so these agents will take the same actions, which implies that, for each of these games, step 2 of the coding algorithm begins at the same history of the game form, which we denote $h^{\emptyset_2}$.

Consider the continuation game starting at $h^{\emptyset_2}$, and recall that h_n^{2*} is the first time an object is clinched in step 2 of game $\Gamma_{n'}$, which is also the first time an object is clinched in step 1 of the continuation game beginning at $h^{\emptyset_2}$. Notice that by the structure of $\succ_n$, the objects $x_{k_{n^{1+1}}}, \dots, x_{k_{n^{2-1}}}$ are lurked at h_n^{2*} in Γ_n , while $x_{k_{n^2}}$ is not, i.e., objects $x_{k_{n^{1+1}}}, \dots, x_{k_{n^{2-1}}}$ are the first lurked objects (in order) along the initial passing path of the game form, beginning at $h^{\emptyset_2}$.

²⁶If $k_{n^{1+1}}$ clinched first in $\Gamma_{n^{1+1}}$ and Γ_n , and is in the same role, then the subcodings $\succ_{n^{1+1}}$ and $\succ_n$ would be the same up to $k_{n^{1+1}}$, which is a contradiction.

The subcodings of games $\Gamma_{n^1+1}, \Gamma_{n^1+2}, \dots, \Gamma_{n^2+1}$ beginning at history $h^{\emptyset 2}$ are:

$$\begin{aligned} & \{i, k_{n^1+1}\} \succ_{n^1+1} \dots \\ & \vdots \\ & k_{n^1+1} \succ_{n^2} k_{n^1+2} \succ_{n^2} \dots \succ_{n^2} k_{n^2-1} \succ_{n^2} \{i, k_{n^2}\} \succ_{n^2} \dots \\ & k_{n^1+1} \succ_{n^2+1} k_{n^1+2} \succ_{n^2+1} \dots \succ_{n^2+1} k_{n^2} \succ_{n^2+1} \{i, k_{n^2+1}\} \dots \end{aligned}$$

By Lemma 20 applied to the continuation game and subcodings beginning at $h^{\emptyset 2}$, in Γ_{n^2+1} , at $h_{n^2+1}^{2*}$, there is an agent ℓ such that ℓ is an active non-lurker at $h_{n^2+1}^{2*}$ that does not move at $h_{n^2+1}^{2*}$ and $x_i \in C_\ell^\varnothing(h_{n^2+1}^{2*})$. Since $\succ_n$ is equivalent to $\succ_{n^2+1}$ up to agent k_{n^2} , and agent k_{n^2} is the last agent in a coding step of game Γ_n , we have that $h_n^{2*} = h_{n^2+1}^{2*}$, by Lemma 4. This implies that at h_n^{2*} , there is an agent ℓ that is an active non-lurker at h_n^{2*} that does not move at h_n^{2*} and $x_i \in C_\ell^\varnothing(h_n^{2*})$ (which may or may not be the same such agent in Γ_{n^2+1} , depending on the role assignment functions).

It remains to show that $\ell \neq k_1, \dots, k_{n^2}, i$. It is clear that $\ell \neq k_1, \dots, k_{n^2}$, since all of these agents are coded by the end of step 2 in Γ_n , while agent ℓ is not. If $\ell = i$, step 3 begins with agent i being offered to clinch x_i . If i is not coded in step 3, then i continues to be active in step 4, which begins with i being offered x_i , etc.. The same continues up to and including step Q , in which i is coded. Since i is coded first in step Q (tying with k_n) x_i is her top object among those that remain at the beginning of step Q . Since $x_i \in C_i(h_n^{(Q-1)*})$, agent i begins step Q by being offered to clinch x_i at the initial history of this step. Since x_i is her top remaining object, she would clinch it, and thus would not tie with k_n , which is a contradiction. Therefore, $\ell \neq i$. This completes the result for $q = 2$.

We then just repeat the arguments for the $q = 2$ case for all $q = 3, 4, \dots, Q-1$, which completes the proof of Claim 1 and hence of Lemma 21. $\blacksquare$

Proof of Lemma 22

The proof strategy is to apply Lemma 21. Thus, consider game Γ_n , and, as in the statement of Lemma 21, let Q be the step of the coding algorithm in which i is coded in game Γ_n and h_n^{1*} be the history at which the first clinching action is taken in step $q \in \{1, \dots, Q\}$ of game Γ_n . If $Q = 1$, we can apply Lemma 21 immediately to conclude i ties with some agent k_{n+1} in $\succ_{n+1}$. So, consider $Q > 1$. To apply Lemma 21 in this case, we must show that at h_n^{1*} , there is an agent ℓ that is an active non-lurker at h_n^{1*} that does not move at h_n^{1*} and $x_i \in C_\ell^\varnothing(h_n^{1*})$.

We first claim that at h_n^{1*} , there are at least $P + 1$ lurked objects. As discussed at the beginning of the proof of Lemma 5, we can assume that $j_1 \succ_n \dots \succ_n j_P \succ_n \dots$ and that the objects $x_{j_1}, \dots, x_{j_P}$ are all lurked at $h_n^{1*} = (\check{h}, a^*, \dots, a^*)$, where $\check{h} = (h_\emptyset, a^*, \dots, a^*)$ is the earliest history at which the youngest lurked object among the x_{j_p} 's, x_{j_P} , becomes lurked. So, there are at least P lurked objects at h_n^{1*} . If there are exactly P lurked objects at h_n^{1*} , then no new object has become lurked on the passing path from $\check{h}$ to h_n^{1*} , which implies that h_n^{1*} is not a terminating history. This follows because a presumption in the statement of the lemma is that x_{k_1} is the $(P + 1)^{th}$ lurked object along the initial passing path, so there must some continuation history $(h_n^{1*}, a^*, \dots, a^*)$ at which x_{k_1} becomes lurked. We also conclude that the agents coded in step 1 of the coding algorithm applied to Γ_n are $j_1, \dots, j_P, k_1$,²⁷ and so one

²⁷This is because the agents coded in a step that is initiated at some h_n^{1*} are: (i) the lurkers at h_n^{1*} , (ii) the non-lurker who moves at h_n^{1*} , and (iii) possibly one more active non-lurker, and if this agent is coded, there must be a tie

of these agents moves at h_n^{1*} . Recall that by Lemma 3, all of these agents receive the same object in all games under consideration, namely x_{j_p} for agent j_p and x_{k_1} for agent k_1 (cf. the last bullet point in the discussion of arms on page 26). Assume first that the agent who moves at h_n^{1*} is some j_p . This means that j_p must be clinching x_{j_p} at h_n^{1*} . As x_{j_p} is a lurked object at h_n^{1*} , Lemma 14 implies that h_n^{1*} is the terminating history, which is a contradiction. So, the agent who moves at h_n^{1*} must be agent k_1 , who clinches x_{k_1} . But then, at h_n^{1*} , x_{k_1} has been offered to an active non-lurker, which implies that x_{k_1} cannot be the next lurked object along the initial passing path (Remark 5), which is again a contradiction.

Therefore, there are at least $P + 1$ lurked object at h_n^{1*} . In particular, the objects $x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n'-1}}$ are all lurked at h_n^{1*} , while $x_{k_{n'}}$ is not, for some $n > n' > 1$.²⁸ Consider game $\Gamma_{n'+1}$ and notice that $\succ_n$ is equivalent to $\succ_{n'+1}$ up to agent $k_{n'}$. By Lemma 4, all agents $j_1, \dots, j_P, k_1, \dots, k_{n'}$ are in the same roles in all of these games, and $h_n^{1*} = h_{n'+1}^{1*}$. By Lemma 20,²⁹ in $\Gamma_{n'+1}$, at $h_{n'+1}^{1*}$, there is an agent ℓ such that ℓ is an active non-lurker at $h_{n'+1}^{1*}$ that does not move at $h_{n'+1}^{1*}$, and $x_i \in C_\ell^\neq(h_{n'+1}^{1*})$. Since $h_n^{1*} = h_{n'+1}^{1*}$, we conclude the same is true at h_n^{1*} . Thus, we can apply Lemma 21 to conclude that agent i must tie with some agent k_{n+1} in $\succ_{n+1}$. ■

Proof of Lemma 23

We begin by showing the result for $n = 1$, as part of the following claim.

CLAIM 1: Assume that there exist σ_0 and σ_1 such that:

$$\begin{aligned} j_1 \cdots j_P \succ_0 i \succ_0 k_1 \succ_0 \cdots \\ j_1 \cdots j_P \succ_1 \{i, k_1\} \succ_1 \cdots \end{aligned}$$

Then:

- (a) We have $h_0^* \not\subseteq h_1^*$, and the agent that moves at h_0^* in Γ_0 is agent i .
- (b) If there exists a σ_2 such that $j_1 \cdots j_P \succ_2 k_1 \succ_2 i \cdots$, then $h_0^* \not\subseteq h_2^*$. Further, in $\succ_2$, agent i must tie with some other agent k_2 .
- (c) If x_{k_1} is not the $(P + 1)^{th}$ lurked object on the initial passing path, then in Γ_2 , agent k_1 clinches at $h_2^* \not\subseteq h_1^*$. Further, at h_2^* , there is an active non-lurker $\ell \neq j_1, \dots, j_P, i, k_1$ such that $x_i \in C_\ell^\neq(h_2^*)$.

The proof of this claim can be found immediately after the proof of this lemma. Now, consider a sequence α such that $n \geq 2$. We will show that i must tie in $\succ_{n+1}$.

In game Γ_n , i is coded in some step of the coding algorithm with some subset of the agents $j_1, \dots, j_P, k_1, \dots, k_{n-1}$. Let Q be the step number in which i is coded in game Γ_n . The goal is to apply Lemma 21, which the following claim allows us to do.

CLAIM 2: If $Q \geq 2$, then at h_n^* , there is an agent ℓ that is an active non-lurker at h_n^* that does not move at h_n^* and $x_i \in C_\ell^\neq(h_n^*)$.

at the end of the step. As there is no tie in $(P + 1)^{th}$ place under $\succ_n$, there is no agent of type (iii), and thus at most $P + 1$ agents are coded, namely the agents $j_1, \dots, j_P, k_1$.

²⁸Because $Q \geq 2$, the last agent coded in step 1 of Γ_n is at most k_{n-1} , which means that $x_{k_{n-1}}$ is not lurked at this point. Thus, the last lurked object is at most $x_{k_{n-2}}$, which implies $n' < n$. That $n' > 1$ follows from the supposition that x_{k_1} is the $(P + 1)^{th}$ lurked object on the initial passing path of the game form, and, as we just showed, at least $P + 1$ objects are lurked at h_n^{1*} .

²⁹In this case, we apply Lemma 20 to the arm $(\sigma_1, \dots, \sigma_{n'+1})$.

The proof of this claim is found below, immediately after the proof of Claim 1. Given Claim 2, we can apply Lemma 21 to conclude that i must tie in $\succ_{n+1}$, and hence proving Claims 1 and 2 is sufficient to complete the proof of Lemma 23.

Proof of Claim 1 (stated above). Since we assume there are at least P lurkers at h_1^* , by the structure of $\succ_1$, there are exactly P lurkers at h_1^* . This implies that the first P lurked objects are $x_{j_1}, \dots, x_{j_P}$. Additionally, objects x_i and x_{k_1} are unlurked at h_1^* , and so x_i and x_{k_1} are agent i and k_1 's favorite objects among the set of those that are unlurked at h_1^* , respectively.

Part (a). Suppose not, then the passing structure of histories implies that $h_1^* \subseteq h_0^*$. Notice that at h_1^* , there must be two active non-lurker roles.

Case 1: $P = 0$. In this case, there are no agents j_p , so at h_1^* , there are exactly two active roles, label them s and s' , and wlog, let $\sigma_1(s) = i$ and $\sigma_1(s') = k_1$. If i clinches at h_1^* in Γ_1 , then $x_i \in C_{s'}^{\mathbb{F}}(h_1^*)$ and $x_i \in C_s(h_1^*)$. Now, for i to be ranked first without ties in $\succ_0$ either (i) x_i is the first lurked object of the game or (ii) i clinches x_i first as an unlurked object, and it has not been offered to another active non-lurker. However, $h_1^* \subseteq h_0^*$ implies that neither (i) nor (ii) can obtain, as x_i has been offered to both active non-lurkers at h_1^* , which is a contradiction.

If k_1 clinches at h_1^* in Γ_1 , then $x_{k_1} \in C_s^{\mathbb{F}}(h_1^*)$ and $x_{k_1} \in C_{s'}(h_1^*)$. Now, $h_1^* \subseteq h_0^*$ implies that in Γ_0 , $\sigma_0^{-1}(k_1) \neq s, s'$.³⁰ Since k_1 is not in either of these roles, there is some $\ell \neq i, k_1$ that is active at h_1^* in Γ_0 and is such that $x_i \in C_{\ell}^{\mathbb{F}}(h_1^*)$. Notice also that since x_{k_1} has been offered to both active agents at h_1^* , it cannot be the second lurked object along the initial passing path (Remark 5), and so for k_1 to be ranked second, there can be at most 3 active agents at h_0^* , in particular agents i, k_1 , and ℓ . If k_1 moves at h_0^* , i must be lurking x_i , and k_1 will tie with agent ℓ . If ℓ moves at h_0^* , it is clear k_1 will not be ranked second without ties. If i moves at h_0^* , then there must be no lurked objects at h_0^* .³¹ But, since $h_1^* \subseteq h_0^*$, we have $x_{k_1} \in C_{\ell}(h_0^*)$, and so, since ℓ was not coded in the first step, she begins the second step by being offered x_{k_1} at the initiating history of the continuation game. Thus, it is impossible for k_1 to be ranked first without ties in this continuation game, a contradiction.

Case 2: $P \geq 1$. In this case, there is at least one lurker j_p at h_1^* . Further, at h_1^* , there are P active lurker roles for the objects $x_{j_1}, \dots, x_{j_P}$, and 2 active non-lurkers roles; label the role that moves at h_1^* as s , and the other active nonlurker at h_1^* as s' . There are three subcases, depending on who is in role s .

Subcase 2.1. $\sigma_1(s) = i$. In this case, we have $\sigma_1(s') = k_1$ and $x_i \in C_{s'}^{\mathbb{F}}(h_1^*)$. We first claim that i cannot be active at h_1^* in Γ_0 . First, notice that i cannot move at h_1^* in Γ_0 , because if she did, she would choose the same action at h_1^* in both games, and would tie in $\succ_0$, just as she did in $\succ_1$. So, $\sigma_0(s) \neq i$. Next, assume i is a lurker at h_1^* in Γ_0 , for some lurked object $x_{j_1}, \dots, x_{j_P}$. Note that x_i cannot be the next object lurked along the initial passing path because it has been offered to (both) active non-lurkers at h_1^* , so at h_0^* , there must be no newly lurked objects, and roles s and s' are still active non-lurkers. The first coding of step Γ_0 thus ends when i clinches x_i , which is unlurked. But, because $h_1^* \subseteq h_0^*$, x_i has been offered to both role s and s' at h_0^* , and one of these is an active non-lurker who does not move at h_0^* , and so i would tie with this agent in $\succ_0$.

Second, assume that $\sigma_0(s') = i$. Then, notice that $x_i \in C_{s'}(h')$ for some $h' \not\subseteq h_1^*$. We claim that i would clinch x_i at this history. Indeed, at h' , role s' is an active non-lurker that is not the

³⁰If $\sigma_0^{-1}(k_1) = s$, then k_1 would clinch at some $h' \not\subseteq h_1^*$; if $\sigma_0^{-1}(k_1) = s'$, then k_1 is in the same role in Γ_0 and Γ_1 , and thus would clinch at $h_0^* = h_1^*$, and would once again tie for first in $\succ_0$.

³¹If there were, it must be x_i . It cannot be lurked by k_1 , since this would mean x_i is her top object, which is a contradiction. So, it must be lurked by some $\ell \neq i, k_1$, and so ℓ will be ranked ahead of or tie with k_1 in $\succ_1$.

terminator.³² This means that only unlurked objects are possible for the agent in this role, and since x_i is i 's favorite unlurked object, she will clinch it at h' , by greedy strategies. Therefore, i is not active at h_1^* in Γ_0 .

Now, i is not active at h_1^* in Γ_0 , but there are two active non-lurkers, those in roles s and s' , and both of these have been offered x_i . Thus, x_i cannot be the next lurked object along the initial passing path of the game form, and so there can be no newly lurked objects at h_0^* . But then, i is not active at h_0^* (since no new agent can become active unless something else becomes lurked), and so i is not coded in this step, which contradicts that she is ranked $(P+1)^{th}$ in $\succ_0$.

Subcase 2.2: $\sigma_1(s) = k_1$. In this case, we have $x_{k_1} \in C_{s'}(h')$ for some $h' \subsetneq h_1^* \subseteq h_0^*$ and $x_{k_1} \in C_s(h_1^*)$. This implies that x_{k_1} cannot be either of the next two lurked objects on the initial passing path of the game form (if they exist). Since k_1 is ordered immediately after i in $\succ_0$ and k_1 does not tie, there can be at most one newly lurked object at h_0^* , and it must be x_i .

We next claim that k_1 cannot be active at h_1^* in Γ_0 . It is clear that $\sigma_0(s) \neq k_1$, because otherwise k_1 would clinch at h_1^* in Γ_0 , and once again tie in $\succ_0$. We also have that $\sigma_0(s') \neq k_1$. To see why, notice that s' is not the terminator role (see footnote 32). So, only unlurked objects are possible for the agent in this role, and thus, if k_1 was in this role, she would clinch x_{k_1} at $h' \subsetneq h_0^*$, since it is her favorite unlurked object. Last, if k_1 lurks some object x_{j_p} at h_1^* , then she strictly prefers x_{j_p} to x_{k_1} . It then must be some agent $j_{p'}$ that moves at h_0^* and clinches a lurked object $x_{j_{p'}}$. This means that $j_{p'}$ is in the terminator role. We claim that $\sigma_0^{-1}(j_{p'}) \neq s, s'$. We know (see footnote 32) that s' is not the terminator role, so $\sigma_0^{-1}(j_{p'}) \neq s'$. If $\sigma_0(s) = j_{p'}$, then s is the terminator role. But, this contradicts that k_1 clinched x_{k_1} first at h_1^* in Γ_1 , since in that game she was in the terminator role and so x_{j_p} is possible for her, and she strictly prefers it. Therefore, in Γ_0 , $j_{p'}$ is in some role s'' that was not active at h_1^* . This implies that one of s or s' is still active at h_0^* in Γ_0 , and whoever it is, this agent has been offered x_{k_1} prior to h_0^* . So, k_1 would tie with this agent in $\succ_0$, a contradiction.

So, k_1 is not active at h_1^* in Γ_0 . So, there is some agent $\ell \neq j_1, \dots, j_P, i, k_1$ that is active at h_1^* in Γ_0 . This agent cannot be a lurker at h_0^* , since if she were, she would necessarily be coded in step 1, and, as x_{k_1} is not lurked at h_0^* , k_1 could at best tie with her. Thus, $\sigma_0^{-1}(\ell) = s$ or s' , and no matter which, we have $x_{k_1} \in C_\ell(h_1^*)$. If x_{k_1} is clinched in step 1, then k_1 can at best tie with ℓ . If k_1 is not coded in step 1, then in at the start of the continuation game for step 2, ℓ is offered x_{k_1} . But, if this is the case, then k_1 cannot be ordered first without ties in step 2, which contradicts the definition of $\succ_0$.

Subcase 2.3: $\sigma_1(s) = j_p$ for some $p = 1, \dots, P$. In this case, agent j_p is clinching a lurked object at h_1^* , and so h_1^* is the terminating history. Then, $h_1^* \subseteq h_0^*$ implies that $h_1^* = h_0^*$. Thus, in Γ_0 , x_i is the first (and only) unlurked object clinched in step 1, and so $x_i \notin C_{s'}(h_1^*)$. So, because there is a tie in Γ_1 , it must be that $x_{k_1} \in C_{s'}(h_1^*)$.

Next, we claim that in Γ_0 , k_1 is not active at h_1^* . Indeed, k_1 is not in role s (as that is occupied by j_p). She also cannot be a lurker, because she is not coded in step 1 (which ends with i). Finally, consider role s' . Notice that s' is not the terminator role (because that is role s), and so, if k_1 were in role s' , she would clinch x_{k_1} at some history $h' \subsetneq h_1^*$ at which it was offered to her, a contradiction.

Therefore, there is some $\ell \neq j_1, \dots, j_P, i, k_1$ that is such that $\sigma_0(s') = \ell$ and $x_{k_1} \in C_\ell(h_1^*)$. Since ℓ is not coded in step 1, she begins the continuation game for step 2 by being offered x_{k_1} . Thus, k_1 cannot be ordered first in step 2 without ties, which is a contradiction.

The above shows that $h_0^* \subsetneq h_1^*$. To finish the proof of part (a), we must show that agent i moves at h_0^* in Γ_0 . Notice that $h_0^* \subsetneq h_1^*$ and the structure of $\succ_1$ implies there can be at most P

³²This follows from Lemma 15. If this role were the terminator, then role s could not be offered x_i at $h_1^* \not\supseteq h'$.

lurkers at h_0^* . First, if there are no lurkers ($P = 0$) at h_0^* , then, it is clear that i must move at h_0^* , as that is the only way she can be ranked first without ties. Now, presume that $P > 0$. If it is some j_p that moves at h_0^* , then j_p clinches a lurked object x_{j_p} , which implies that h_0^* is the terminating history, which contradicts $h_0^* \not\sqsubseteq h_1^*$. Therefore, no agent $j_1, \dots, j_P$ can move at h_0^* . Since there can be at most P lurkers at h_0^* , given that i is ranked $(P + 1)^{th}$ without tying, the only other possibility is that it is agent i that moves at h_0^* and clinches x_i .

Part (b). We first show that $h_0^* \not\sqsubseteq h_2^*$. By part (a), $h_0^* \not\sqsubseteq h_1^*$. This means that agent i cannot move at h_0^* in Γ_1 . Nor can any potential agent j_p , because if they did, they would be clinching a lurked object, which means h_0^* is the terminating history, which contradicts $h_0^* \not\sqsubseteq h_1^*$. Therefore, it must be k_1 that moves at h_0^* in Γ_1 .

By way of contradiction suppose that $h_0^* \not\sqsubseteq h_2^*$ fails; because of the passing structure of this histories, it means that $h_2^* \subseteq h_0^*$. The structure of $\succ_2$ implies that k_1 clinches at h_2^* in Γ_2 , which also means that h_2^* and h_0^* are controlled by different roles, and further $h_2^* \not\sqsubseteq h_0^*$.³³ So, in Γ_0 , it must be some agent $\ell \neq j_1, \dots, j_P, i, k_1$ that moves at h_2^* . But then, we have $x_{k_1} \in C_\ell(h_0^*)$, so at the initial history of the continuation game that begins step 2, agent ℓ is offered x_{k_1} , and so k_1 cannot be ordered first in step 2, which is a contradiction to the definition of Γ_0 . Therefore, $h_0^* \not\sqsubseteq h_2^*$.

Thus, we have $h_0^* \not\sqsubseteq h_1^*, h_2^*$, and so agent i does not move at h_0^* in Γ_1 or Γ_2 .

Case 1: Agent k_1 moves at h_0^* in Γ_2 . Here, k_1 is in the same role as in Γ_1 , and so $h_1^* \not\sqsubseteq h_2^*$. This implies that i must clinch at h_1^* in Γ_1 , and so i does not move at h_1^* in Γ_2 . If some j_p moves at h_1^* in Γ_2 , then this agent must also clinch at h_2^* , and she must clinch a lurked object. This means that i must be a lurker for some x_{j_p} , and so she strictly prefers x_{j_p} to x_i . But then, the agent that moves at h_1^* is in the terminator role, and so in Γ_1 , i is in the terminator role, and since she clinches x_i at h_1^* , this implies that x_i is her top object (lurked or unlurked) by Lemma 15(ii), which is a contradiction. So, it must be some $\ell \neq j_1, \dots, j_P, i, k_1$ that moves at h_1^* in Γ_2 , and so $x_i \in C_\ell^{\neq}(h_2^*)$ in Γ_2 . Since ℓ is not coded in step 1, she is offered x_i at the initial history of the continuation game that begins step 2. Therefore, i cannot be ranked first without ties in this continuation game.

Case 2: Some agent $j_1, \dots, j_P$ moves at h_0^* in Γ_2 . This agent, say j_p , must be the one clinching at h_2^* (since j_p is not a lurker at h_0^* , but ultimately receives a lurked object), and she must clinch a lurked object. This implies that the agent who moves at h_0^* is in the terminator role, and that h_2^* is the terminating history, so $h_1^* \subseteq h_2^*$. Let r be the other role that is active at h_1^* . Since there is a tie in $\succ_1$, this role must be such that either $x_i \in C_r^{\subseteq}(h_1^*)$ or $x_{k_1} \in C_r^{\subseteq}(h_1^*)$. In the latter subcase, x_{k_1} cannot be the next lurked object along the passing path (from h_1^*), and so there must be no newly lurked objects at h_2^* . Next, notice that $\sigma_2(r) \neq k_1$, because otherwise, k_1 would clinch x_{k_1} at the history $h' \not\sqsubseteq h_1^*$ where it was offered in Γ_2 . Thus, k_1 can at best tie with the agent $\sigma_2(r)$, which is a contradiction.

For the subcase $x_i \in C_r^{\subseteq}(h_1^*)$, if $\sigma_2(r) = k_1$, then there is some agent $\ell \neq j_1, \dots, j_P, k_1$ who is a lurker for some $x_{j_1}, \dots, x_{j_P}$. We also have $\ell \neq i$. This is because the agent who moves at h_0^* is in the terminator role, and so in Γ_0 , i is in this role, and since she clinches, x_i is her top available object (lurked or unlurked), and therefore i cannot lurk any of the x_{j_p} 's. Therefore, agent ℓ will be ranked ahead of i in $\succ_2$, a contradiction.³⁴ We also cannot have $\sigma_2(r) = i$, because i would clinch x_i at the history $h' \not\sqsubseteq h_1^*$ at which she was offered x_i . Thus, $\sigma_2(r) = \ell$ for some $\ell \neq j_1, \dots, j_P, i, k_1$. Agent ℓ is not coded in step 1, and thus, she is offered x_i at the initiating history of the continuation game that begins step 2, and so i cannot be ranked first without tying in step 2.

³³If they were the same role, then k_1 is in this role in Γ_1 , and would clinch at h_2^* in Γ_1 , which is a contradiction.

³⁴Note that x_i cannot be lurked at h_2^* , since it has been offered to agent j_p at h_0^* , who is the terminator.

Part (c). If x_{k_1} is not the $(P+1)^{th}$ lurked object, then, because k_1 is ordered without tying in $\succ_2$, at h_2^* , k_1 must clinch x_{k_1} , and it has not been offered to another active non-lurker. Notice also that $h_0^* \not\subseteq h_2^*$ implies that i does not move at h_0^* in Γ_1 or Γ_2 , and that k_1 moves at h_0^* in Γ_1 . If k_1 moves at h_0^* in Γ_2 , then she is in the same role in both games, and so $h_1^* \not\subseteq h_2^*$. This also means that i moves at h_1^* in Γ_1 (because if it was k_1 , then x_{k_1} is offered to both active roles at h_1^* , and so in Γ_2 , k_1 would clinch at some $h' \not\subseteq h_2^*$). Thus, x_i has been offered to both active non-lurker roles at h_1^* . This implies that i cannot be active at h_1^* in Γ_2 , and so there is some $\ell \neq j_1, \dots, j_P, i, k_1$ such that $x_i \in C_\ell(h_2^*)$ in Γ_2 . If k_1 does not move at h_0^* in Γ_2 , then it is some $\ell \neq j_1, \dots, j_P, i, k_1$ that moves at h_0^* . In either case, we have $x_i \in C_\ell^{\subseteq}(h_2^*)$ in Γ_2 , establishing Claim 1. ■

Proof of Claim 2 (stated above). Since it is without loss of generality to assume that there are at least P lurkers at h_n^* , there are two cases. Recall that k_1 is ranked strictly, without ties, in $\succ_n$.

Case 1: There are exactly P lurkers at h_n^* . In this case, k_1 is the last agent coded in step 1 of Γ_n . Consider game Γ_2 , and notice that $\succ_n = \succ_2$ up to agent k_1 . Since agent k_1 is the last agent in a coding step, by Lemma 4, all agents $j_1, \dots, j_P, k_1$ are in the same roles in Γ_2 and Γ_n , and $h_n^* = h_2^*$. Further, notice that x_{k_1} is not the $(P+1)^{th}$ lurked object along the initial passing path,³⁵ and so, by Claim 1 part (c), there is an agent ℓ that is an active non-lurker at h_2^* that does not move at h_2^* and $x_i \in C_\ell^{\subseteq}(h_2^*)$. Since $h_2^* = h_n^*$, the result holds.

Case 2: There are strictly greater than P lurkers at h_n^* . In this case, the objects $x_{j_1}, \dots, x_{j_P}, x_{k_1}, \dots, x_{k_{n'-1}}$ are lurked at h_n^* , while $x_{k_{n'}}$ is not, where $n > n' > 1$.³⁶ Consider game $\Gamma_{n'+1}$, and notice that $\succ_n$ is equivalent to $\succ_{n'+1}$ up to agent $k_{n'}$. Therefore, by Lemma 4, all agents $k_1, \dots, k_{n'}$ are in the same roles in all of these games, and $h_n^* = h_{n'+1}^*$. By Lemma 20, in $\Gamma_{n'+1}$, at $h_{n'+1}^*$, there is an active agent ℓ such that ℓ is an active non-lurker at $h_{n'+1}^*$ that does not move at $h_{n'+1}^*$ and $x_i \in C_\ell^{\subseteq}(h_{n'+1}^*)$. Since $h_n^* = h_{n'+1}^*$, Claim 2 holds and so does Lemma 23. ■

Proof of Lemma 24

By the assumption that $n, m \geq 2$ in α and α' , we have that there exist (at least) the following codings:

$$\begin{aligned} j_1 \cdots j_P &\succ_1 \{i, k_1\} \succ_1 \cdots \\ j_1 \cdots j_P &\succ_2 k_1 \succ_2 \{i, k_2\} \cdots \\ j_1 \cdots j_P &\succ'_2 i \succ'_2 \{k_1, k'_2\} \cdots \end{aligned}$$

We start by presenting the following two conditions, one of which, when combined with prior lemmas, will imply that Statement 1 of the lemma holds, and the other of which will imply Statement 2 of the lemma holds.

- Condition 2: In Γ_2 , at h_2^* there is an active non-lurker ℓ such that ℓ does not move at h_2^* and $x_i \in C_\ell^{\subseteq}(h_2^*)$.

³⁵If k_1 clinches at h_n^* , then x_{k_1} is offered to an active non-lurker, and so cannot be the next lurked object along the initial passing path; if some j_p clinches at h_n^* , then they are clinching a lurked object, and so h_n^* is the terminating history, which again implies that x_{k_1} is not $(P+1)^{th}$ lurked object along the initial passing path (because no such object exists).

³⁶Because $Q \geq 2$, the last agent coded in step 1 of Γ_n is at most k_{n-1} , which means that $x_{k_{n-1}}$ is not lurked, i.e., the last lurked object is at most $x_{k_{n-2}}$, which is why we have $n' < n$.

- Condition 2': In Γ'_2 , at $h_{2'}^*$, there is an active non-lurker ℓ such that ℓ does not move at $h_{2'}^*$ and $x_{k_1} \in C_\ell^{\neq}(h_{2'}^*)$.³⁷

We first show that these conditions imply the lemma. Then, we show that one of these conditions must hold.

We will show that Condition 2 implies that Statement 1 of Lemma 24 holds. The two statements are symmetric, so this will also show that Condition 2' implies Statement 2 of Lemma 24.

To show Condition 2 implies Statement 1, we use Lemma 21. So, consider the sequence of codings

$$\begin{aligned}
 j_1 \cdots j_P \succ_1 \{i, k_1\} \succ_1 \cdots \\
 j_1 \cdots j_P \succ_2 k_1 \succ_2 \{i, k_2\} \cdots \\
 j_1 \cdots j_P \succ_3 k_1 \succ_3 k_2 \succ_3 \{i, k_3\} \succ_3 \cdots \\
 \vdots \\
 j_1 \cdots j_P \succ_n k_1 \succ_n k_2 \succ_n k_3 \succ_n \cdots \succ_n k_{n-1} \succ_n \{i, k_n\} \succ_n \cdots \\
 j_1 \cdots j_P \succ_{n+1} k_1 \succ_{n+1} k_2 \succ_{n+1} k_3 \succ_{n+1} \cdots \succ_{n+1} k_{n-1} \succ_{n+1} k_n \succ_{n+1} i \cdots
 \end{aligned}$$

Recall that it is wlog to assume that there are at least P lurked objects at $h_{n'}^*$ for each n' . We claim further that in this case, there are exactly P lurked objects at $h_{n'}^*$ for each n' . For $n' = 1$, this follows from the fact that i and k_1 tie. For $n' > 1$, the next ordered agent is k_1 . So, if there were $p > P$ lurked objects at $h_{n'}^*$, the $(p+1)^{th}$ lurked object would have to be x_{k_1} , which contradicts the supposition of the lemma. Therefore, for all $n' = 1, \dots, n+1$, at $h_{n'}^*$ in game $\Gamma_{n'}$, there are exactly P lurked objects, and by definition, these must be $x_{j_1}, \dots, x_{j_P}$, in this order.

Next, notice that for all $n' \geq 2$, since there are exactly P lurked objects at $h_{n'}^*$, the set of agents coded in step 1 of $\Gamma_{n'}$ must be $j_1, \dots, j_P, k_1$. In particular, this is true for Γ_2 and Γ_n , and since $\succ_2$ is equivalent to $\succ_n$ up to agent k_1 , by Lemma 4, all of these agents are in the same roles in both games, and $h_n^{1*} = h_2^*$. By Condition 2, there is some agent ℓ such that ℓ is an active non-lurker that does not move at h_2^* and $x_i \in C_\ell^{\neq}(h_2^*)$. Since $h_n^{1*} = h_2^*$, we have that in Γ_n , there is some agent ℓ' that is an active non-lurker at h_n^{1*} and that does not move at h_n^{1*} and $x_i \in C_{\ell'}^{\neq}(h_n^{1*})$. Further, $Q \geq 2$. Thus, all of the conditions of Lemma 21 are satisfied, and we conclude that i must tie with some agent k_{n+1} in $\succ_{n+1}$.

We complete the proof of Lemma 24 by showing that at least one of Condition 2 or Condition 2' must hold. This assertion is proven as Lemma 27 below.

LEMMA 27: *Assume that there are three codings:*

$$\begin{aligned}
 j_1 \cdots j_P \succ_A \{i, k\} \succ_A \cdots \\
 j_1 \cdots j_P \succ_B i \succ_B \cdots \\
 j_1 \cdots j_P \succ_C k \succ_C \cdots
 \end{aligned}$$

such that:

- *At each of h_A^*, h_B^*, h_C^* , the objects $x_{j_1}, \dots, x_{j_P}$ are all lurked, in this order, and*

³⁷We use $h_{2'}^*$ (instead of h_2^{1*}) to denote the first history at which an object is clinched in game Γ'_2 (under role assignment σ'_2).

• Neither x_i nor x_k are the $(P+1)^{th}$ lurked object on the initial passing path of the game. Then, one of the following conditions must hold:

Condition (B): In Γ_B , at h_B^* there is an active non-lurker ℓ such that ℓ does not move at h_B^* and $x_k \in C_\ell^{\subseteq}(h_B^*)$.

Condition (C): In Γ_C , at h_C^* , there is an active non-lurker ℓ such that ℓ does not move at h_C^* and $x_i \in C_\ell^{\subseteq}(h_C^*)$.

Proof of Lemma 27. First, notice that in each of the games, there must be exactly P lurkers at h_γ^* for $\gamma = A, B, C$. It is a presumption of the lemma that there are at least P lurkers. To see that there are at most P lurkers, notice that, for Γ_A , this holds because i and k tie. In Γ_B , it holds because x_i is not the next lurked object along the initial passing path, and thus, x_i must be the first—and since there is no tie, only—unlurked object that is coded in step 1. The same applies to Γ_C . Therefore, in Γ_A , there are exactly $P+2$ agents coded in step 1, while in Γ_B and Γ_C , there are exactly $P+1$ agents coded in step 1.

In Γ_A , at h_A^* , there are P active lurker roles and two active non-lurker roles. The objects $x_{j_1}, \dots, x_{j_P}$ are lurked, and x_i and x_k are unlurked. Let s be the active non-lurker role that moves at h_A^* , and s' the role of the other active non-lurker. One of x_i or x_k must be the first unlurked object that is clinched in step 1 of the coding algorithm, either at h_A^* itself, or in the chain of assignments that follows. Assume it is x_i (a symmetric argument works if it is x_k). This implies that $x_i \in C_{s'}^{\subseteq}(h_A^*)$, and $\sigma_A(s') = k$. There are two cases, depending on who is in role s .

Case 1: $\sigma_A(s) = j_p$ for some p . Agent j_p must be clinching a lurked object at h_A^* , which implies that h_A^* is the terminating history, and s is the terminator role. This means that s' is not the terminator role, and so $x_k \notin C_{s'}^{\subseteq}(h_A^*)$; indeed, if this were true, then agent k would have clinched it in Γ_A , because it is her favorite unlurked object and only unlurked objects are possible for a non-lurker who is not the terminator (Lemma 15(iv)). It also means that agent i must be a lurker for some object $x_{j_{\bar{p}}}$, and thus, agent i strictly prefers $x_{j_{\bar{p}}}$ to x_i .

Now, consider game Γ_C . The agents coded in step 1 of Γ_C are $j_1, \dots, j_P, k$, and so it must be one of these agents that moves at h_C^* .

Subcase 1.1: The agent that clinches at h_C^* is some $j_{p'}$. Here, h_C^* must also be the terminating history, and so $\sigma_C(s) = j_{p'}$ and $h_A^* = h_C^*$. Since k is coded in step 1, she must then be a lurker, and so there is some other agent $\ell \neq j_1, \dots, j_P, k$ such that $\sigma_C(s') = \ell$. We claim that $\ell \neq i$. Indeed, if $\ell = i$, then there is some history $h' \subsetneq h_C^*$ such that $x_i \in C_i(h')$. Since s' is not the terminator role, only unlurked objects are possible for i in Γ_C , and since x_i is her top unlurked object, she would clinch at h' , a contradiction. Therefore, $\sigma_C(s') = \ell \neq i$, and Condition (C) holds.

Subcase 1.2: Agent k clinches at h_C^* in Γ_C . Here, we have $\sigma_C(s) = k$, because, as we saw above, $x_k \notin C_{s'}^{\subseteq}(h_A^*)$ and h_A^* is the terminating history, so $h_C^* \subseteq h_A^*$. Let $h' \subsetneq h_A^*$ be the history at which role s' is offered to clinch x_i .

If $h_C^* \not\supseteq h'$, then, by similar logic to subcase 1.1, $\sigma_C(s') = \ell$ for some $\ell \neq j_1, \dots, j_P, k, i$, and Condition (C) holds.

Finally, consider $h_C^* \subsetneq h'$.³⁸ In Γ_B , since there are exactly $P+1$ agents coded in step 1, x_i is the first (and only) unlurked object that is clinched, and since there is no tie, it has not been offered to another active non-lurker. This implies that $h_B^* \subseteq h' \subsetneq h_A^*$. Since h_B^* is not the terminating history, it must be an unlurked object that is clinched, and therefore, it must be i that clinches x_i . If $\sigma_B(s) = i$, then i is in the terminator role, and would not clinch x_i first

³⁸Note that $h_C^* = h'$ is ruled out because role s' moves at h' , while role s moves at h_C^* .

at h_B^* (recall that she prefers $x_{j_{\bar{p}}}$ to x_i). Thus, it must be that $\sigma_B(s') = i$, and i clinches x_i at h_B^* . If $h_B^* \not\subseteq h_C^*$, then by similar logic to the above, Condition (C) holds. If $h_C^* \not\subseteq h_B^*$, then $x_k \in C_s^{\subseteq}(h_B^*)$ for the agent in role s . Notice that $\sigma_B(s) \neq k$, because if so, then k has the same roles in Γ_B and Γ_C , and so would clinch at $h_C^* \not\subseteq h_B^*$ in Γ_B , a contradiction. It is also immediate that $\sigma_B(s) \neq j_1, \dots, j_P$, since they must be in the lurker roles for their respective objects. Thus, $\sigma_B(s) = \ell$ for some $\ell \neq j_1, \dots, j_P, i, k$, and Condition (B) holds.

Case 2: $\sigma_A(s) = i$. We once again have that role s' is not the terminator role,³⁹ and so, as in Case 1, $x_k \notin C_{s'}^{\subseteq}(h_A^*)$. Once again, consider game Γ_C . As in Case 1, there are two subcases.

Subcase 2.1: The agent that clinches at h_C^* in Γ_C is some $j_{p'}$. Here, $j_{p'}$ clinches a lurked object at h_C^* , and so h_C^* is the terminating history. This implies that $h_A^* \subseteq h_C^*$, and $\sigma_C(s) = j_{p'}$. But then, notice that the agent in role s' is an active non-lurker at h_C^* that does not move at h_C^* , and $x_i \in C_{s'}^{\subseteq}(h_C^*)$. Since this agent is not coded in step 1, we know that $\sigma_C(s') \neq j_1, \dots, j_P, k$. If $\sigma_C(s') = i$, then i is offered to clinch x_i at some $h' \not\subseteq h_C^*$, and since s' is not the terminator role, only unlurked objects are possible for her, and therefore, since x_i is i 's top object, she would clinch at h' , a contradiction. Thus, $\sigma_C(s') = \ell$ for some $\ell \neq j_1, \dots, j_P, i, k$, and $x_i \in C_{\ell}^{\subseteq}(h_C^*)$, i.e., Condition (C) holds.

Subcase 2.2: The agent that clinches at h_C^* in Γ_C is k . Since k clinches first, and x_k is unlurked, all lurked objects are immediately assigned to their lurkers, which implies that j_p is in the lurker role for x_{j_p} for all $p = 1, \dots, P$.

If $h_A^* \subseteq h_C^*$, then, at h_C^* , there are two active non-lurkers, $\sigma_C(s)$ and $\sigma_C(s')$, and both have been offered x_i . One of these must be k . If $\sigma_C(s') = k$, then notice that $\sigma_C(s) \neq i$, because if $\sigma_C(s) = i$, then i is in the same role in Γ_A and Γ_C , and would clinch at h_A^* in Γ_C , which contradicts that k clinches first in Γ_C . Thus, $\sigma_C(s) = \ell \neq i$. If $\sigma_C(s) = k$, then if $\sigma_C(s') = i$, then i is in the non-terminator role, and $x_i \in C_i(\tilde{h})$ for some $\tilde{h} \not\subseteq h_A^* \subseteq h_C^*$, and since x_i is i 's favorite unlurked object, she will clinch it at $\tilde{h}$, a contradiction. Therefore, in either case, there is some agent $\ell \neq j_1, \dots, j_P, i, k$ such that $x_i \in C_{\ell}^{\subseteq}(h_C^*)$, and Condition (C) holds.

It remains to consider $h_C^* \not\subseteq h_A^*$. Here, we must have $\sigma_C(s) = k$, because if $\sigma_C(s') = k$, then as we showed above, $x_k \notin C_{s'}^{\subseteq}(h_A^*)$, which contradicts that k clinches at h_C^* . Now, consider Γ_B . In Γ_B , since there are exactly $P + 1$ agents coded in step 1, x_i is the first (and only) unlurked object that is clinched, and the agents coded in step 1 are $j_1, \dots, j_P, i$.

If $h_B^* \subseteq h_C^*$, then, $h_B^* \not\subseteq h_A^*$, and h_B^* is not the terminating history. Thus, in Γ_B , agent i must move at h_B^* and clinch x_i . This implies that $\sigma_B(s') = i$, because if $\sigma_B(s) = i$, then i has the same role in Γ_A and Γ_B and clinches at both h_B^* and h_A^* , which contradicts that $h_B^* \not\subseteq h_A^*$. Further, this means that $h_B^* \neq h_C^*$, because role s moves at h_C^* and role s' moves at h_B^* . Thus, at h_C^* in Γ_C , we have $x_i \in C_{s'}^{\subseteq}(h_C^*)$. We cannot have $\sigma_C(s') = i$, because i would clinch at h_B^* in Γ_C , a contradiction. Therefore, $\sigma_C(s') = \ell$ for some $\ell \neq j_1, \dots, j_P, i, k$ and $x_i \in C_{\ell}^{\subseteq}(h_C^*)$, and thus, Condition (C) holds.

If $h_C^* \not\subseteq h_B^*$, then if some $j_{p'}$ clinches at h_B^* in Γ_B , then h_B^* is the terminating history, and $h_A^* \subseteq h_B^*$. But then, there is an active non-lurker—the agent $\sigma_B(s')$ —that has been offered to clinch x_i prior to h_B^* , and so i would at best tie with this agent in $\succ_B$, a contradiction. Thus, it must be i that clinches at h_B^* in Γ_B , which implies that $\sigma_B^{-1}(i) = s$ or s' . If $\sigma_B^{-1}(i) = s$, then i has the same roles in Γ_A and Γ_B , and so $h_A^* = h_B^*$, and i would tie with the agent in role s' in $\succ_B$, a contradiction. Thus, $\sigma_B(s') = i$. This means that h_C^* and h_B^* are controlled by different roles, and $x_k \in C_s^{\subseteq}(h_B^*)$. Finally, we cannot have $\sigma_B(s) = k$, because then k is in the same role as Γ_C , and would clinch at $h_C^* \not\subseteq h_B^*$ in Γ_B . So, we must have $\sigma_B(s) = \ell$ for some $\ell \neq j_1, \dots, j_P, i, k$, and in Γ_B , $x_k \in C_{\ell}^{\subseteq}(h_B^*)$. Therefore, Condition (B) holds.

³⁹This follows from Lemma 15.

Finally, notice that all of this was done under the assumption that x_i was the first unlurked object that was clinched in step 1 of the coding algorithm in Γ_A . The other possibility is that this object is x_k . However, everything is symmetric, and so the exact same argument, swapping the i and k , shows that either Condition (B) or Condition (C) must hold in this case as well. ■

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